Ecole polytechnique
federale de Lausanne



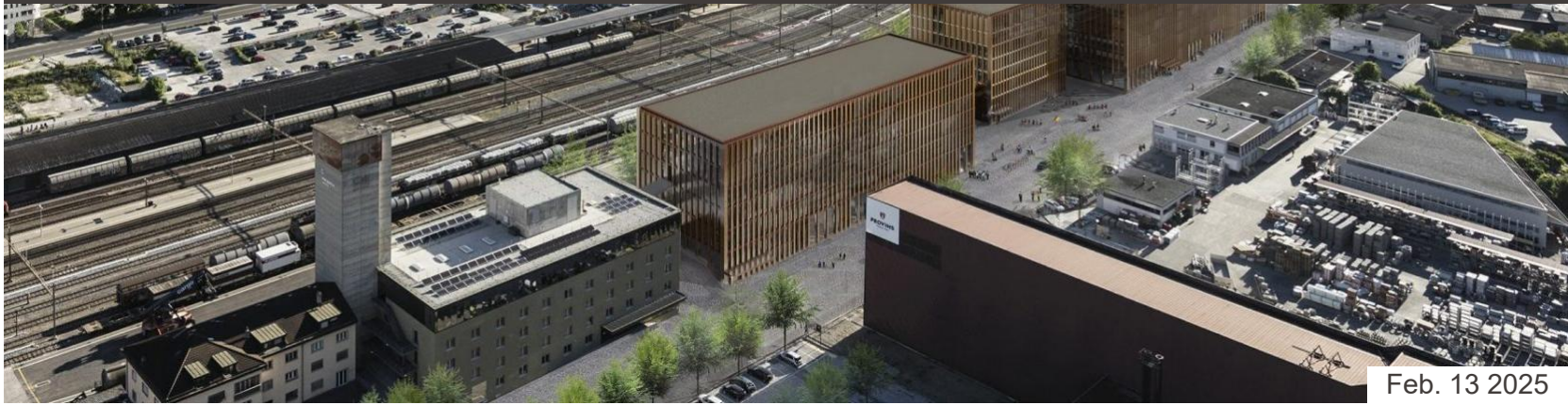
February 2026 intra-task participants meeting

Industrial Energy-Related Technologies and Systems iets
An IEA Technology Collaboration Programme

Task XXIV Process Integration for Industry Decarbonization

Subtask 1 - Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics

IETS Task XXIV Team
Daniel Florez-Orrego, Oktay Boztas, Meire Ribeiro Domingos, Pullah Bhatnagar, François Maréchal



Feb. 13 2025



Content

- Recap of the Task XXIV – 1 year
- Template (re)definition
- Rules of the game
- Database infrastructure
- Annual report(s) Y2025
- Linkedin page/ IETS website/ Forum
- Next steps
- AOB



Recap of the Task XXIV

Objectives

- **Define** the ontologies, specifications and procedures to generate, validate, and share models.
- **Share** knowledge and innovations in industrial processes and decarbonization technologies.
- **Discuss** impact on energy systems models, distribution networks and seasonal demands (**ETSAP**)
- **Compare** process integration methods/tools for ranking decarbonization pathways (metrics)
- **Share** experiences and case studies to outline decarbonization roadmaps and symbiosis.
- **Disseminate** through workshops, papers, scientific events, and educational programs.

Structure and term

Time frame: 01.01.2025 – 01.11.2027

Task managers:

- Daniel Florez-Orrego
- François Maréchal

Process Integration for Industry Decarbonization

Recent Developments

The proposal for this new Task was approved by the IETS ExCo on 20 November 2024. The Task will start early 2025.

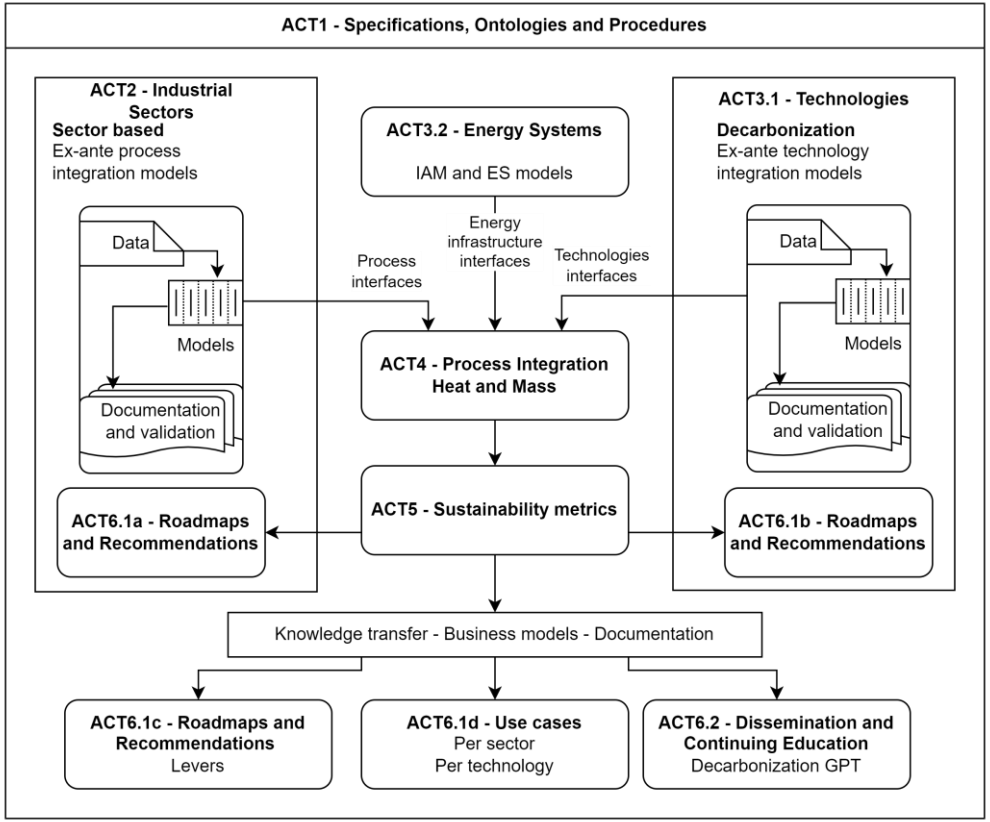
Read the Task Proposal

TASK XXIV

START DATE 2025-01-01

END DATE 2027-12-31

Read the task proposal **HERE**



EPFL Team +

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Colombia



Brazil



Turkey



India

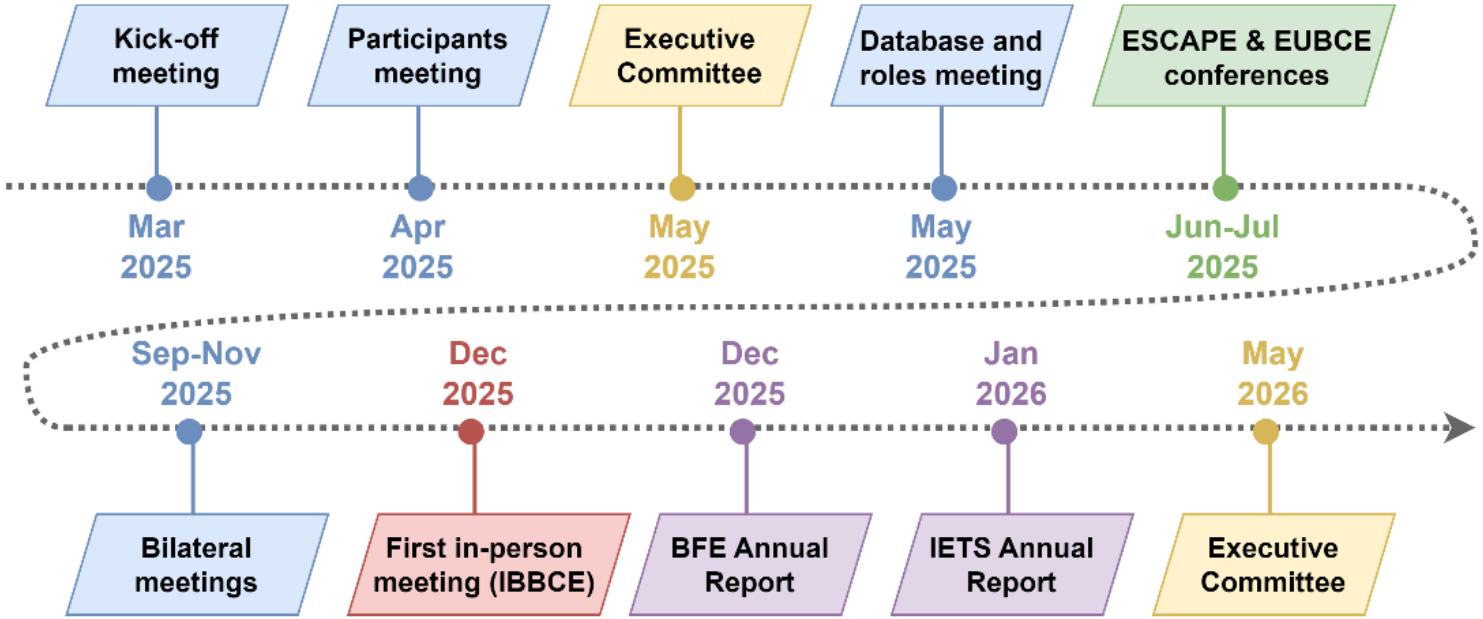




Switzerland

Belgium

Recap of the Task XXIV





Bilateral and annual intra-task meetings

Credit

- Citation, co-authorship arrangements, commercialization

Data management

- Findability, accessibility, interoperability, reproducibility

Anonymization

- Sensitive industrial data into shareable formats



Software

- Dominated by commercial simulators, complemented by open-source and in-house platforms

Barriers

- Financial, time, and legal constraints; Industrial partner objections → anonymization

Tiered access

- Open access vs. confidentiality → agreements, embargo periods, regulations

Country	Key Point
Switzerland	Maximization of open access data for public information
Sweden	Impact of project confidentiality on sharing capabilities
Netherlands	Requirement for open sharing of proprietary software models
Italy	Model portability across scales (engineering, chemistry, control)
Germany	Data anonymization while maintaining real-world relevance
Denmark	Added value of documentation beyond journals publication
Chile	Data exchange policies between researchers and industry
Canada	Sustainability metrics for industrial symbiosis
Portugal	Validation by sub-teams then by the broader community
Belgium	Different interfaces for various stakeholder types
Colombia	Educational platform as a model database dissemination



Rules of the game

Credits and attributions

- **Main expectations:** Develop new research collaborations from database utilization, use models in industrial related projects, disseminate existing and new models.

Save time: Avoid reinventing the wheel

- **Attribution and recognition** (depending on the levels of model use):

- Citation: Mandatory for all model usage. Citations in publications and studies that uses the data.
- Co-authorship: Negotiated case-by-case (**central**, **adapted**, **ancillary use**).
- Commercial purposes: less interested.

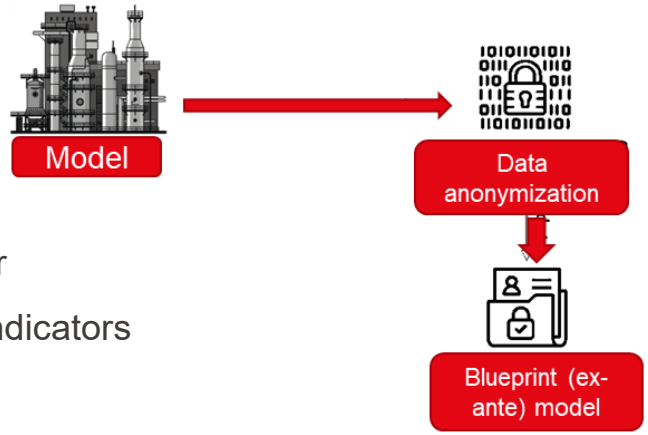
- **Licensing Options:**

- License: Creative Commons (CC-BY, CC-BY-SA), MIT, or custom licenses.
- Embargo periods: aligned with journal publication timelines.

Anonymization and validation process

❑ Anonymization

- Normalization to reference capacity (per unit of product)
- Representative or synthetic data based on real obfuscated + thermodynamic data
- Aggregation: Uncertainty ranges instead of absolute values



❑ Validation & Review

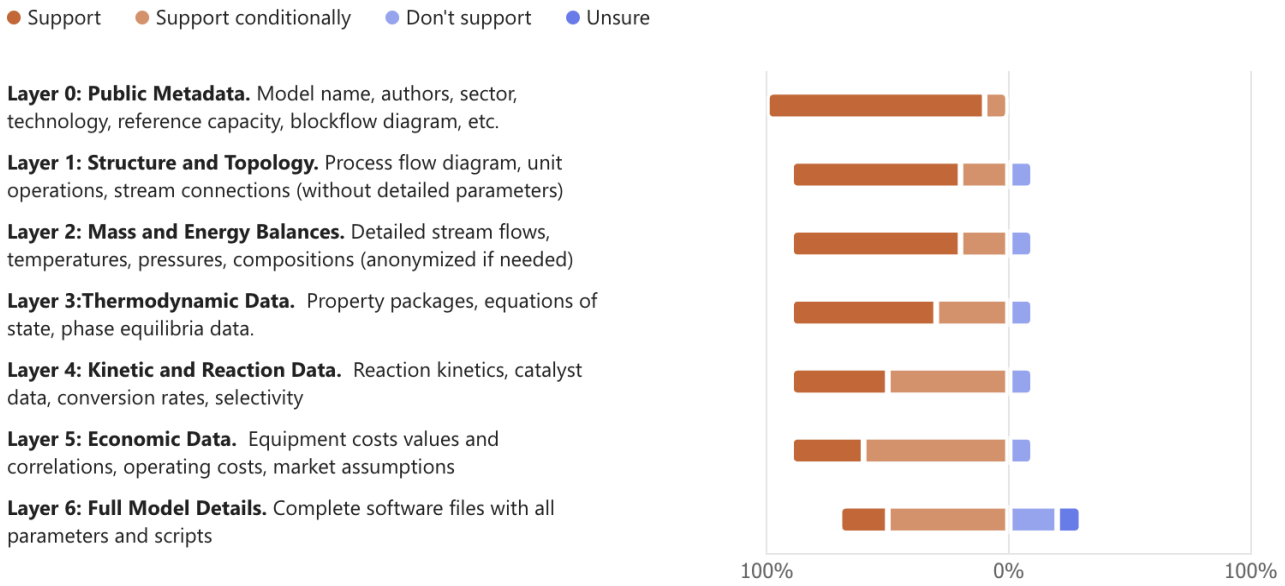
- Specialized sub-teams for sector/technology-specific validation
- Sufficient information required for peer review at designated layer
- Maintain validated and non-validated versions with clear status indicators

Improve technical quality through standardization and peer-review

Confidentiality considerations

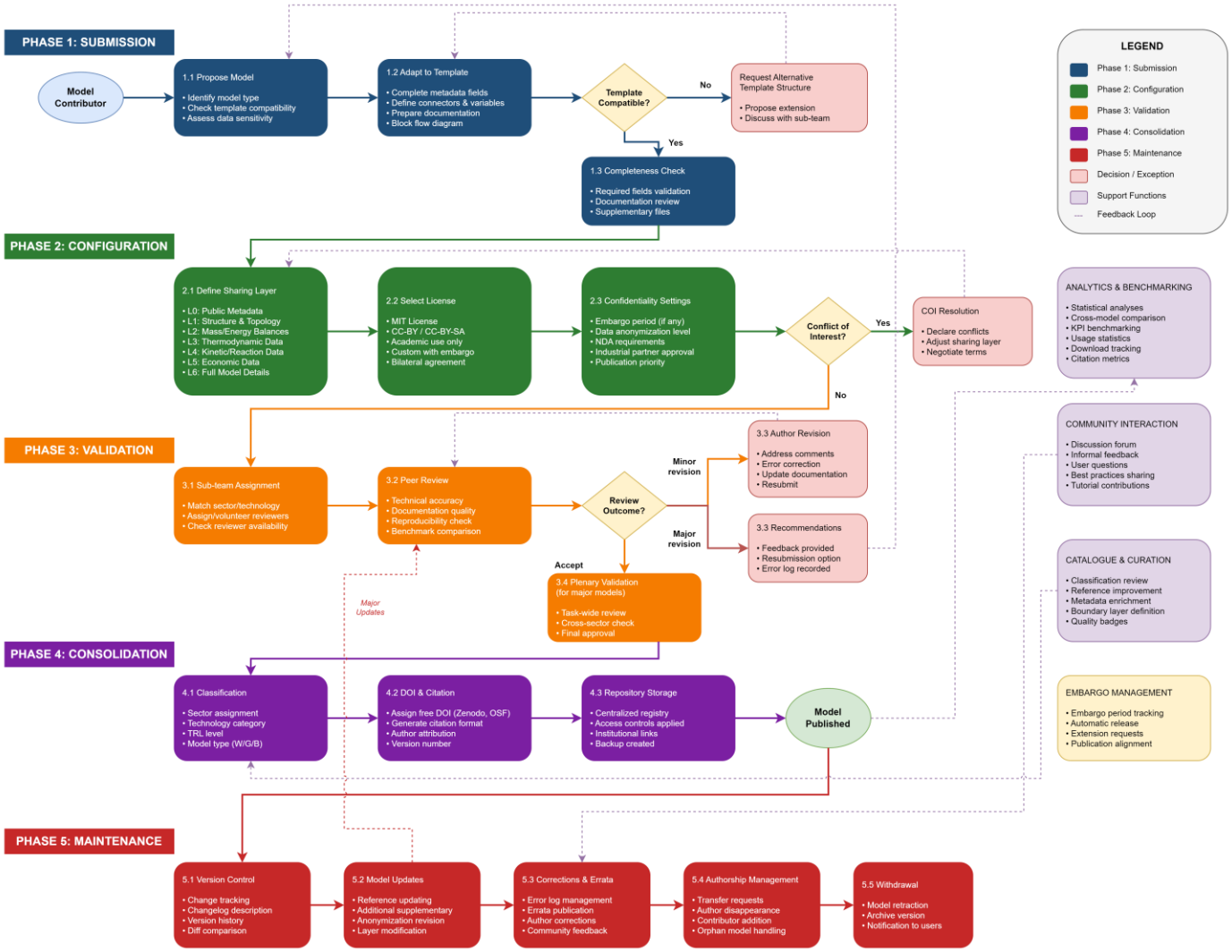
Tiered model sharing framework: Define sharing layer based on data sensitivity and institutional constraints

- **Layer 0–2** (Metadata, Structure, Mass/Energy Balances) → Openly accessible
- **Layer 3–4** (Thermodynamic, Kinetic Data) → Conditional access
- **Layer 5–6** (Economic Data, Full Model) → Bilateral agreement



Sharing procedure

1. **PHASE 1: Submission**
   - Submit model adapted to standard template
   - Complete required fields: metadata, connectors, variables, documentation
2. **PHASE 2: Configuration**
   - Specify sharing layer and granularity, select license (MIT, CC, embargo, etc.)
   - Attach supplementary materials
3. **PHASE 3: Validation**
   - Submit for sub-team peer review based on sector/technology
   - Provide sufficient information for review at specified layer
4. **PHASE 4 and 5: Consolidation and maintenance**
   - Agree to openly share at designated layer
   - Model stored in centralized repository with access and version controls


PHASE 1: SUBMISSION
Model Contributor
1.1 Propose Model
• Identify model type
• Check template compatibility
• Assess data sensitivity
1.2 Adapt to Template
• Complete metadata fields
• Define connectors & variables
• Prepare documentation
• Block flow diagram
Template Compatible?
No
Request Alternative Template Structure
• Propose extension
• Discuss with sub-team
Yes
1.3 Completeness Check
• Required fields validation
• Documentation review
• Supplementary files
PHASE 2: CONFIGURATION
2.1 Define Sharing Layer
• L0: Public Metadata
• L1: Structure & Topology
• L2: Mass/Energy Balances
• L3: Thermodynamic Data
• L4: Kinetic/Reaction Data
• L5: Economic Data
• L6: Full Model Details
2.2 Select License
• MIT License
• CC-BY / CC-BY-SA
• Academic use only
• Custom with embargo
• Bilateral agreement
2.3 Confidentiality Settings
• Embargo period (if any)
• Data anonymization level
• NDA requirements
• Industrial partner approval
• Publication priority
Conflict of Interest?
Yes
COI Resolution
• Declare conflicts
• Adjust sharing layer
• Negotiate terms
No
PHASE 3: VALIDATION
3.1 Sub-team Assignment
• Match sector/technology
• Assign/volunteer reviewers
• Check reviewer availability
3.2 Peer Review
• Technical accuracy
• Documentation quality
• Reproducibility check
• Benchmark comparison
Review Outcome?
Minor revision
3.3 Author Revision
• Address comments
• Error correction
• Update documentation
• Resubmit
Major revision
3.3 Recommendations
• Feedback provided
• Resubmission option
• Error log recorded
Accept
3.4 Plenary Validation (for major models)
• Task-wide review
• Cross-sector check
• Final approval
Major Updates
PHASE 4: CONSOLIDATION
4.1 Classification
• Sector assignment
• Technology category
• TRL level
• Model type (W/G/B)
4.2 DOI & Citation
• Assign free DOI (Zenodo, OSF)
• Generate citation format
• Author attribution
• Version number
4.3 Repository Storage
• Centralized registry
• Access controls applied
• Institutional links
• Backup created
Model Published
PHASE 5: MAINTENANCE
5.1 Version Control
• Change tracking
• Changelog description
• Version history
• Diff comparison
5.2 Model Updates
• Reference updating
• Additional supplementary
• Anonymization revision
• Layer modification
5.3 Corrections & Errata
• Error log management
• Errata publication
• Author corrections
• Community feedback
5.4 Authorship Management
• Transfer requests
• Author disappearance
• Contributor addition
• Orphan model handling
5.5 Withdrawal
• Model retraction
• Archive version
• Notification to users
LEGEND
Phase 1: Submission
Phase 2: Configuration
Phase 3: Validation
Phase 4: Consolidation
Phase 5: Maintenance
Decision / Exception
Support Functions
Feedback Loop
ANALYTICS & BENCHMARKING
• Statistical analyses
• Cross-model comparison
• KPI benchmarking
• Usage statistics
• Download tracking
• Citation metrics
COMMUNITY INTERACTION
• Discussion forum
• Informal feedback
• User questions
• Best practices sharing
• Tutorial contributions
CATALOGUE & CURATION
• Classification review
• Reference improvement
• Metadata enrichment
• Boundary layer definition
• Quality badges
EMBARGO MANAGEMENT
• Embargo period tracking
• Automatic release
• Extension requests
• Publication alignment

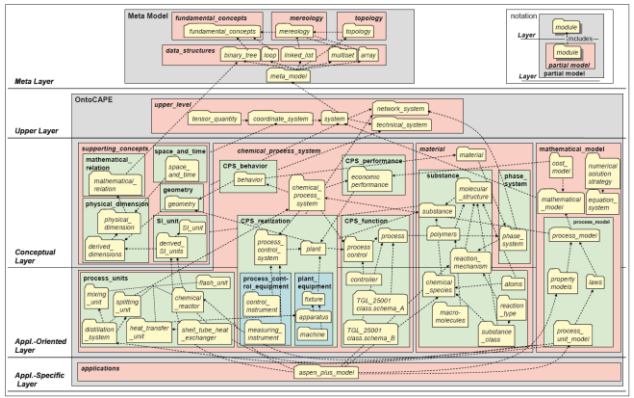


Template feedback

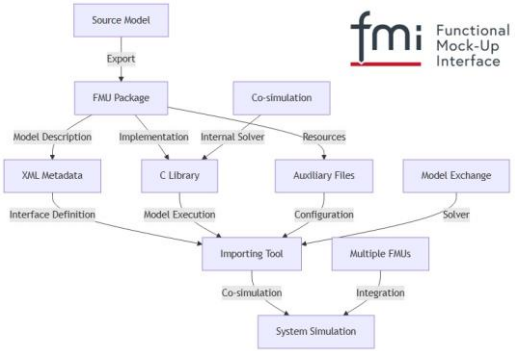
Model structure

- Support of a simplified framework

ONTOCAPE



FMI



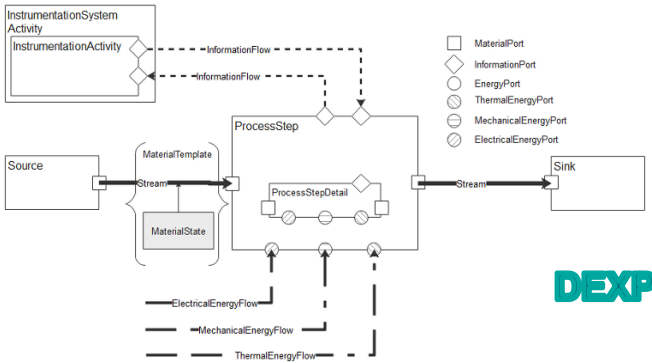
fmi Functional Mock-Up Interface

```
<?xml version="1.0" encoding="ISO-8859-1"?>
<UnitOperation
          xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
          xsi:noNamespaceSchemaLocation="GenericMoTUnit.xsd">

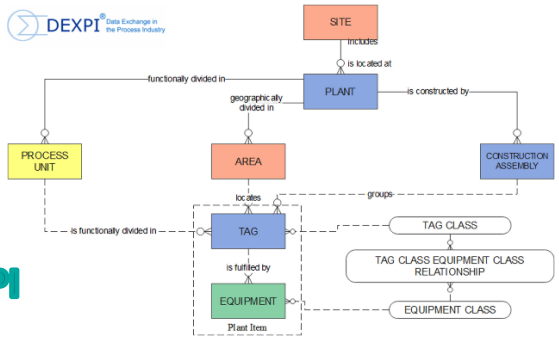
 <Globals>
   <MOTFile>C:\Program Files\COMoTUO\FuelGen.mot</MOTFile>
   <Name>FuelCell</Name>
   <Description>Test for CAPE-OPEN ProSim-Mot</Description>
   <MinimumCompoundsNumber>1</MinimumCompoundsNumber>
   <MaximumCompoundsNumber>2</MaximumCompoundsNumber>
   <MinimumInputMaterialPortsNumber>1</MinimumInputMaterialPortsNumber>
   <MaximumInputMaterialPortsNumber>1</MaximumInputMaterialPortsNumber>
   <MinimumOutputMaterialPortsNumber>1</MinimumOutputMaterialPortsNumber>
   <MaximumOutputMaterialPortsNumber>2</MaximumOutputMaterialPortsNumber>
 </Globals>

 <UserParameters>
   <UserParameter>
     <Name>Residence time</Name>
     <Description>Mean Residence time</Description>
     <Mode>input</Mode>
     <Type>real</Type>
     <DefaultValue>500.0</DefaultValue>
     <LowerBound>0.0</LowerBound>
     <UpperBound>1000.0</UpperBound>
     <MOTVariable>tau</MOTVariable>
   </UserParameter>
   ...
```

CAPEOPEN



DEXPI

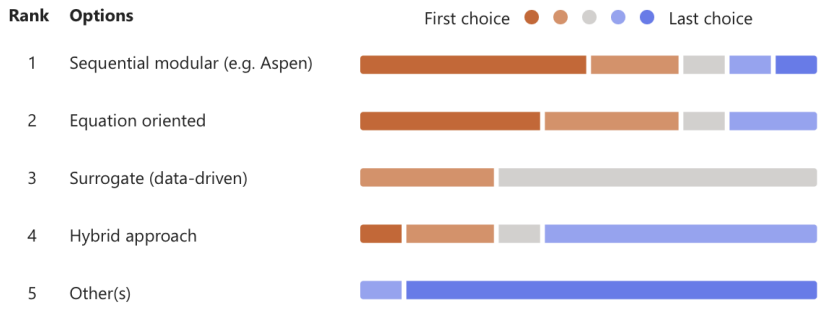


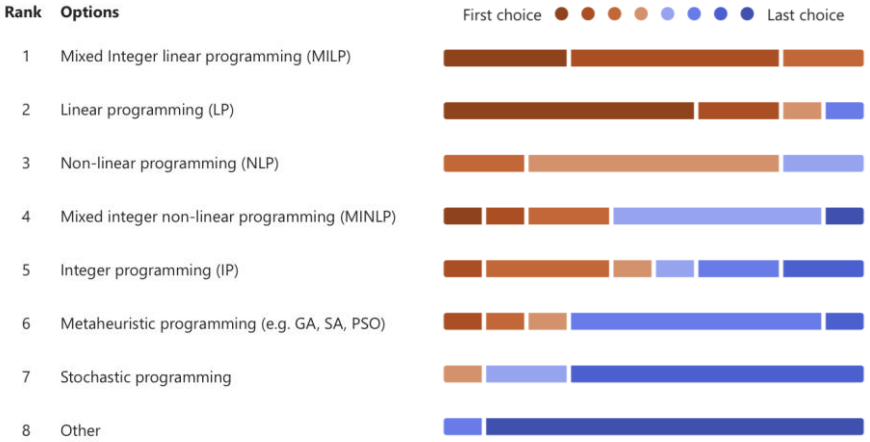
ISO

Refs. [1-4]

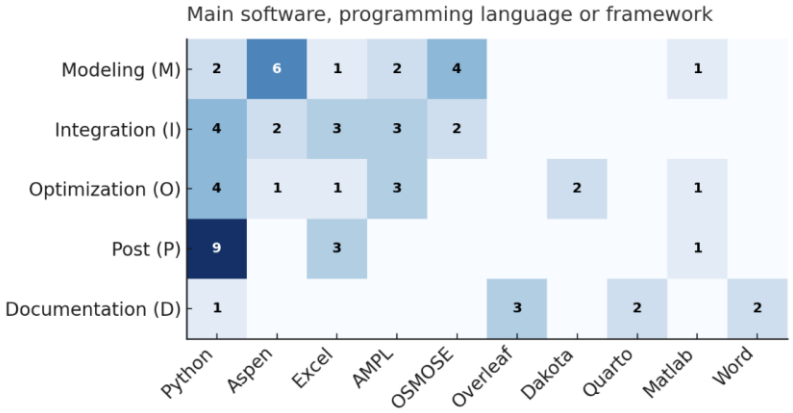
Essentials tools for the participants

Low familiarity with complex ontos and standards



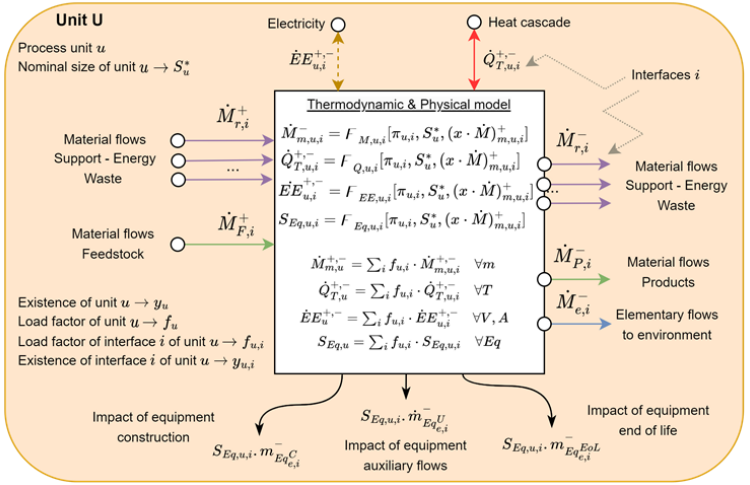


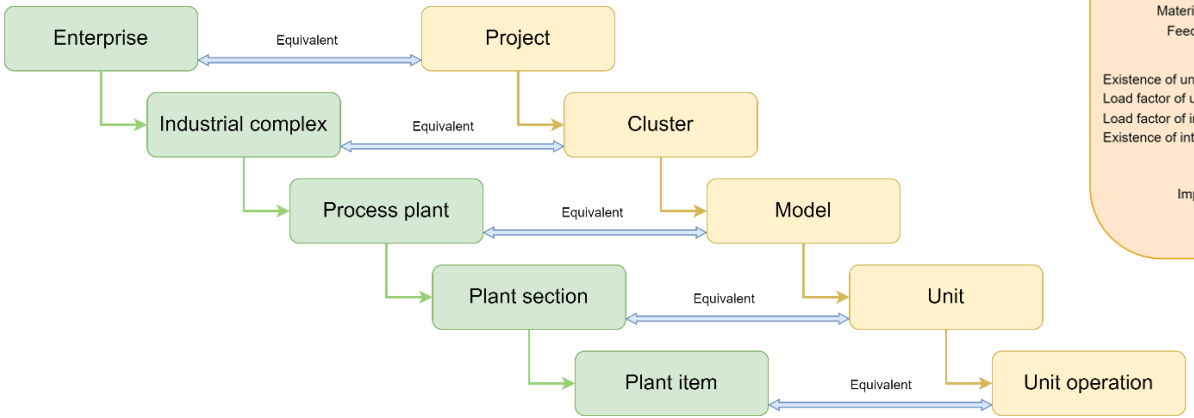
- **Most common tools:** Aspen/ MILP integration + Python for pre/post-processing (data handling, KPIs visualization, and wrappers).
- **Excel** still widely used for interfacing and calculator blocks.
- **Documentation in** Obsidian/Quarto/Word/Overleaf



Model context, scope and structure

- Model Structure:
  - Metadata: authorship, software, version, level of sharing, etc.
  - Connectors: mass, energy, emissions, and information flows with quality → to be associated to KPI coefficients.
  - Main decision variables (input parameters) and derived (calculated) variables.
  - Documented (referenced) parameters and assumptions
  - Simplified block flow diagram
  - Supplementary material (flowsheeting, publications, other files)



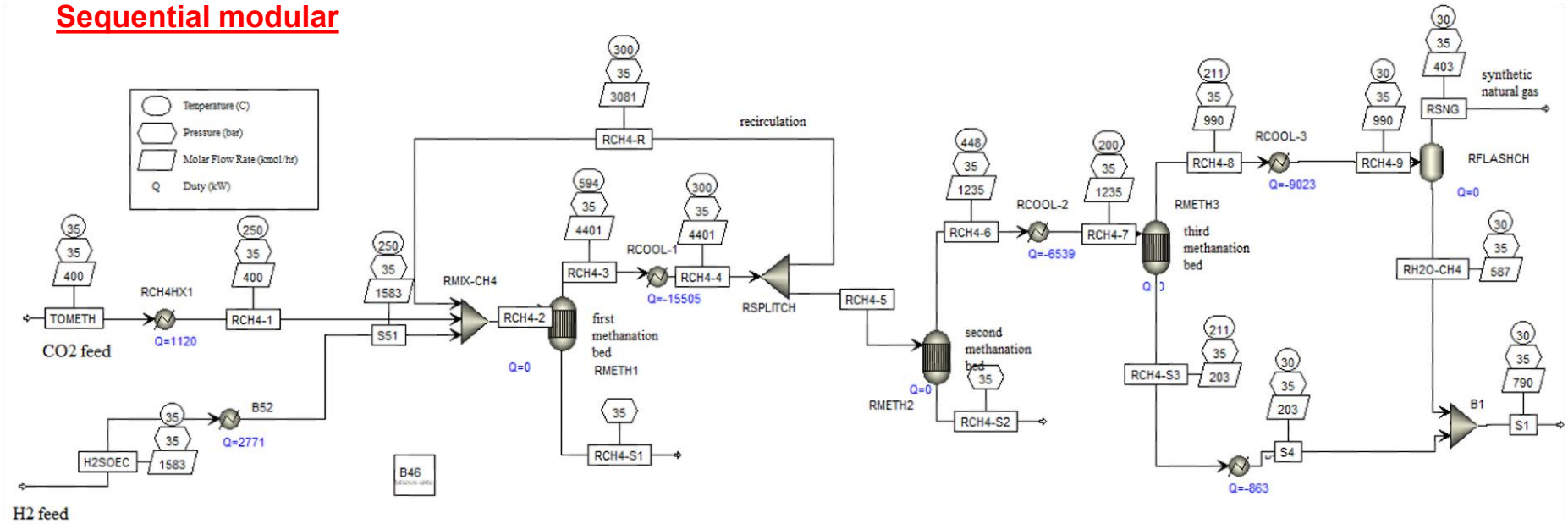


Plant breakdown structure compatible with ISO 10209 and adopted by DEXPI

Model interfaces definition

- Example of a simplified methanator model in Aspen Plus (TREMP™):

Sequential modular



Exothermic, multiple bed catalytic reactor with intercooling, it consumes CO_2 and H_2 producing CH4 and H2O considering the equilibrium of reactions of Sabatier and Water Gas Shift

Model interfaces definition

1. MODEL HIERARCHY

SYSTEM (e.g., Power-to-Gas Plant)

MODEL 1
Electrolyzer
- SOEC units
- Heat recovery
- Power conditioning

H_2

MODEL 2
Methanator
- Reactors (RMETH1-3)
- Coolers (RCOOL1-3)
- Separators, mixers

Hierarchy:

SYSTEM → Contains multiple MODELs
MODEL → Contains UNIT OPERATIONS
UNIT OP → Processes STREAMS
STREAMS → Characterized by VARIABLES

Power | H_2O | CO_2 feed | SNG

2. UNIT OPERATIONS (Nodes)

HEAT TRANSFER
- Heaters
- Coolers
- Heat exchangers
- Condensers
- Reboilers
- Economizers

REACTION
- CSTR
- PFR
- Equilibrium
- Stoichiometric
- Yield
- Electrolyzers

SEPARATION
- Flash drums
- Distillation columns
- Absorbers
- Membranes
- Cyclones
- Filters

FLOW HANDLING
- Mixers
- Splitters
- Compressors
- Pumps
- Expanders
- Valves
- Pipes

Generate

3. STREAM TYPES FOR PROCESS INTEGRATION

VAR - KEY VARIABLES

QT Streams:
- T_{in}: Inlet temperature (°C)
- T_{out}: Outlet temperature (°C)
- Q: Heat duty (kW): (+) heating required, (-) cooling required
- ΔT_{mh}: Min. approach temp (K)

EE Streams:
- P: Power (kW)
- V: Voltage (kV)

RS Streams:
- ṁ: Mass flow (kg/s)
- T: Temperature (°C)
- P: Pressure (bar)
- x: Composition (mol/mass %)

QT - HEAT STREAMS (Quality: Temperature)

Sources: Heaters, coolers, exo/endothermic reactors, condensers, reboilers, heat exchangers

Hot streams: Release heat (cooling required)
Cold streams: Absorb heat (heating required)

Variables: T_{in}, T_{out}, Q (kW), ΔT_{mh}, Cp×ṁ

EE - POWER STREAMS (Electricity/Mechanical)

Sources: Compressors, pumps, expanders, electrolyzers, motors, generators, electrical heaters

Variables: P (kW), V (voltage), efficiency, load factor

RS - CONNECTORS (Resources)

Types:
- Feedstock (raw materials)
- Products (main outputs)
- Byproducts (secondary outputs)
- Utilities (water, steam, air)
- Wastes (emissions, effluents)
- Information (control signals)

Variables: ṁ (kg/s), composition, T, P, phase

4. CONNECTOR CLASSIFICATION

EXTERNAL CONNECTORS

Boundary streams connecting model to external world

- May be UNMATCHED (only IN or only OUT)
- Define model boundaries
- Interface with infrastructure or external models.

Examples:
IN: CO_2 feed, H_2 feed, Power
OUT: SNG product, Waste heat

INTERNAL CONNECTORS

Streams connecting unit operations within model

- Always MATCHED (IN-OUT pairs)
- Close mass/energy balances
- Enable internal heat integration

Examples:
RCH4-1 → RCH4-2

Recirculation loops
Inter-stage streams

EQUIPMENT LIST

Unit ID	Type	Description
RCH4HX1	Heater	CO_2 preheater
B52	Heater	H_2 preheater
RMIX-CH4	Mixer	Feed mixer
RMETH1	Reactor	1st methanation
RMETH2	Reactor	2nd methanation
RMETH3	Reactor	3rd methanation
RCOOL-1	Cooler	Inter-stage cool 1
RCOOL-2	Cooler	Inter-stage cool 2
RCOOL-3	Cooler	Final cooler
RSPLITCH	Splitter	Recycle split
RFLASHCH	Separator	Product flash

5. DATA EXTRACTION WORKFLOW

5.1 Flowsheet Analysis
Import process flowsheet (Aspen, DWSIM, etc.)

→ 5.2 Unit Recognition
Identify heaters, coolers, reactors, separators, etc.

→ 5.3 QT Extraction
Extract heat duties, temperatures from thermal units

→ 5.4 Connector Identification
Map external & internal stream connections

→ 5.5 Variable Assignment
Define T, P, ṁ, composition for each stream

→ 5.6 Table Generation
Populate QT, EE, RS tables for database

6. DATABASE TABLES

HEAT STREAMS TABLE (QT)

Stream ID	Unit	Type	T_{in} (°C)	T_{out} (°C)	Q (kW)	ΔT_{mh} (K)
QT-001	RCH4HX1	Cold	35	250	+1120	10
QT-002	RCOOL-1	Hot	594	300	-15505	5
QT-003	RMETH1	Hot	250	594	-8500	15
...						

POWER CONNECTORS TABLE (EE)

Connector ID	Unit	Direction	P (kW)	V (kV)	Scope
EE-001	H2SOEC	IN	5000	0.4	External
EE-002	COMP-1	IN	150	0.4	External
...					

MASS CONNECTORS TABLE (RS)

Conn. ID	Stream	Dir.	Type	ṁ (kg/s)	T (°C)	P (bar)	Composition	Scope
RS-001	CO2feed	IN	Feedstock	4.89	35	35	100% CO_2	Ext
RS-002	H2feed	IN	Feedstock	0.44	35	35	100% H_2	Ext
RS-003	RSNG	OUT	Product	2.75	35	35	95% CH_4	Ext
...								

Model interfaces definition

Note that **alternative interfaces for the "heat cascade" exposition** can be considered:

- Reaction enthalpy
  - Heat leaving as exothermic reaction enthalpy at the reactor intercooling temperature.
  - Heat leaving in the form of steam flow rate at specified pressure (and temperature) .
  - Heat leaving at the lowest temperature available in the cooling tower.

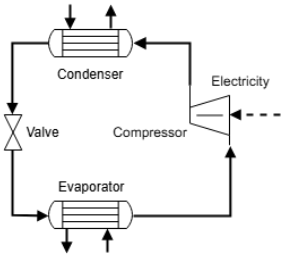
- Reactants preheating
  - Reactants preheating expressed as a cold process stream.
  - Reactants preheating coming in the form of electricity (Joule-Thomson).
  - Reactants preheating at the highest temperature considered for a fired heater.

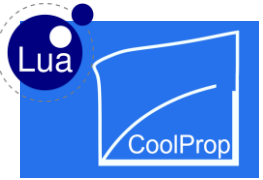
- Produced water leaving either at low or high temperature.

- No waste heat recovery available at all (only mass interfaces).

All those assumptions and data availability affect the characterization and the integrability of the models when seen from an external user.

Model interfaces definition





Equation oriented

```
---------------
--PARAMETERS --
---------------

et.inputs = {
    -- Evaporator temperatures
    Evap_Tin = {default=20, unit='°C'},-- Evaporator temperature inlet
    Evap_Tout = {default=20, unit='°C'}, -- Evaporator temperature outlet
    -- Condenser temperatures
    Cond_Tin = {default=110, unit='°C'}, -- Condenser temperature inlet
    Cond_Tout = {default=110, unit='°C'}, -- Condenser temperature outlet
    Cond_Qmax = {default=10000, unit='kW'},-- Condenser reference capacity
    exeff = {default = 0.5,unit = ''}, -- Lorentz efficiency (empirical)
    -- dtmin contribution
    dtmin_2ph={default = 2,unit = 'C'},

---------------
-- ECONOMICS ---
---------------

    n   = {default = 40, unit = 'year', description = 'Lifetime'},
    i = {default = 0.06, unit = '-', description = 'Discount interest rate'},
    CEPCI_2020   = {default = 600.2,    unit = '-', description = 'actual CEPCI'},
    CEPCI_2008   = {default = 575.4,    unit = '-', description = 'CEPCI 2008'},

}

---------------
--CALCULATED---
---------------

et.outputs = {
    -- COP calculation
    COPcarnot = {unit='kW',job='(Cond_Tin+273)/(Cond_Tin-Evap_Tin)' },
    COP = {unit='kW',job='exeff*COPcarnot()' },
    -- Electricity consumption
    W_heatpump = {unit='kW',job='Cond_Qmax/COP()' },
    -- Evaporator
    Evap_Qmax = {unit='kW',job='Cond_Qmax*(COP()-1)/COP()' },
```

```
---------------
-- ECONOMICS ---
---------------
Annuity = {job = '(i*(i+1)^n)/((i+1)^n-1)',unit = '-'},

Cinv2_HP = {unit='kW', job=
    function()
        if Cond_Qmax < 500 then
            HP_Cinv2= (750*Cond_Qmax)*(CEPCI_2020/CEPCI_2008)*Annuity()
        else
            HP_Cinv2= (300*Cond_Qmax)*(CEPCI_2020/CEPCI_2008)*Annuity()
        end
    return HP_Cinv2
    end
 }

}

---------------
---- LAYERS ---
---------------

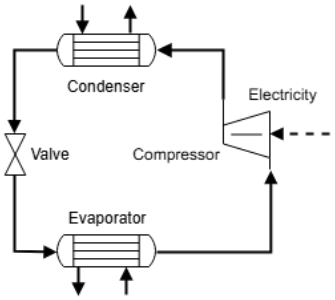
et:addLayers {Elec = {type= 'ResourceBalance', unit = 'kW'} }

---------------
----- UNITS ---
---------------

et:addUnit('HeatpumpforSOEC',{type='Utility', Fmin = 0, Fmax = 1, Cost1=0, Cost2=0,
Cinv1=0, Cinv2='Cinv2_HP', Impact1=0, Impact2=0})
et['HeatpumpforSOEC']:addStreams{
    -- Heat streams
    qt_evap = qt({tin = 'Evap_Tin', hin = 0, tout='Evap_Tout', hout='Evap_Qmax',
dtmin='dtmin_2ph', alpha=1}),
    qt_cond = qt({tin = 'Cond_Tin', hin = 'Cond_Qmax', tout='Cond_Tout', hout=0,
dtmin='dtmin_2ph', alpha=1}),
    -- Electricity consumption
    elec_in = rs({'Elec', 'in', 'W_heatpump'})
}
```

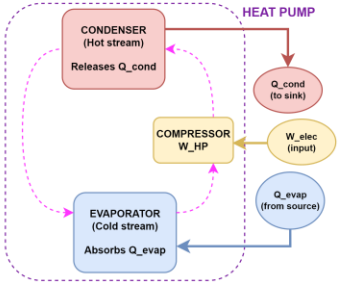
Model interfaces definition

Equation oriented



Lua CoolProp

1. HEAT PUMP MODEL STRUCTURE



2. INPUT PARAMETERS

```
THERMAL PARAMETERS:

Evap_Tin = 20 [°C] Evaporator inlet
Evap_Tout = 20 [°C] Evaporator outlet
Cond_Tin = 110 [°C] Condenser inlet
Cond_Tout = 110 [°C] Condenser outlet
Cond_Qmax = 10000 [kW] Heat pumping duty
exeff = 0.5 [-] Lorentz efficiency

ΔT_min CONTRIBUTIONS:

dtmin_2ph = 2 [K] Two-phase

ECONOMIC PARAMETERS:

n = 40 [yr] Lifetime
i = 0.06 [-] Discount rate
CEPCI_2026 = 596.2 [-] Cost index 2020
CEPCI_2008 = 575.4 [-] Cost index 2008
```

3. THERMODYNAMIC EQUATIONS

Carnot COP (theoretical maximum):

$$COP_{Carnot} = (T_{cond} + 273) / (T_{cond} - T_{evap})$$

Real COP (with exergy efficiency):

$$COP = \eta_{ex} \times COP_{Carnot}$$

Electrical power consumption:

$$W_{HP} = Q_{cond} / COP$$

Evaporator heat duty:

$$Q_{evap} = Q_{cond} \times (COP - 1) / COP$$

4. ECONOMIC EQUATIONS

Annuity factor:

$$A = i \times (1+i)^n / ((1+i)^n - 1)$$

Investment cost (annualized):

$$C_{inv} = c_{spec} \times Q_{cond} \times (CEPCI_{2026}/CEPCI_{2008}) \times A$$

where:

$c_{spec} = 750$ €/kW if $Q < 500$ kW

$c_{spec} = 300$ €/kW if $Q \geq 500$ kW

5. CALCULATED VALUES (Example)

```
Given:
T_evap=20°C, T_cond=110°C, Q_cond=10000kW, η_ex=0.5

COP_Carnot = (110+273)/(110-20) = 4.26
COP = 0.5 × 4.26 = 2.13
W_HP = 10000 / 2.13 = 4695 kW
Q_evap = 10000 × (2.13-1)/2.13 = 5305 kW

Annuity = 0.06×(1.06)^40/((1.06)^40-1) = 0.0665

C_inv = 300 × 10000 × (596.2/575.4) × 0.0665
= 206,600 €/year
```

6. STREAM DEFINITION MAPPING

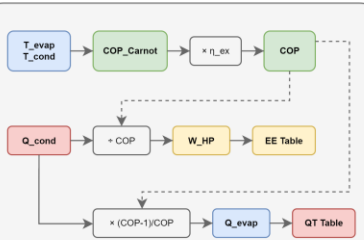
```
QT STREAM DEFINITION:

qt_evap = qt({
        tin = Evap_Tin, -- 20°C
        hin = 0, -- enthalpy at inlet
        tout = Evap_Tout, -- 20°C
        hout = Evap_Qmax, -- 5305 kW (absorbed)
        dtmin = dtmin_2ph, -- 2 K
        alpha = 1 -- heat transfer coeff
})

qt_cond(
        tin = Cond_Tin, -- 110°C
        hin = Cond_Qmax, -- 10000 kW (released)
        tout = Cond_Tout, -- 110°C
        hout = 0,
        dtmin = dtmin_2ph, -- 2 K
        alpha = 1
})


EE STREAM DEFINITION:
rs(Layer, Direction, Flow)
        elec_in = rs({
        'Elec', -- Layer name
        'in', -- Direction
        'W_heatpump' -- 4695 kW
})
```

7. DATA EXTRACTION



8. DATABASE TABLES

Heat Streams Table (QT)

Stream	Type	T_in(°C)	T_out(°C)	Q(kW)	ΔT_min (K)
qt_evap	Cold	20	20	+5305	2
qt_cond	Hot	110	110	-10000	2

Sign convention: (+) heat absorbed, (-) heat released

Note: Two-phase streams → constant temperature (T_in = T_out)
ΔT_min = dtmin_2ph for phase-change processes

Power Connectors Table (EE)

Stream	Dir	Layer	P(kW)	Description
elec_in	IN	Elec	4695	Compressor power

Layer type: ResourceBalance (electricity grid connection)

Economics Table

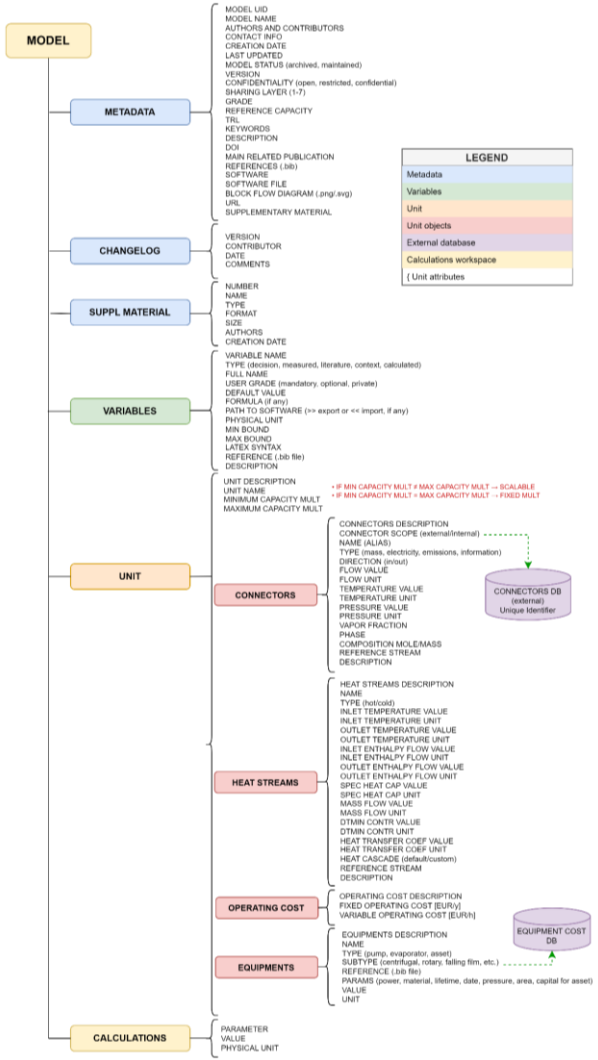
Parameter	Value	Unit	Description
Cinv1	0	€/y	Fixed investment
Cinv2	206,600	€/y	Variable investment
Cost1	0	€/y	Fixed operating
Cost2	0	€/y	Variable operating
Impact1	0	-	Fixed environmental
Impact2	0	-	Variable environmental

Fmin 0 - Minimum load factor
Fmax 1 - Maximum load factor

Model relations tree

- ✓ Independent of paradigm:
  - Equation oriented
  - Sequential modular simulation
  - Response surface or surrogate
  - Black box (single point)

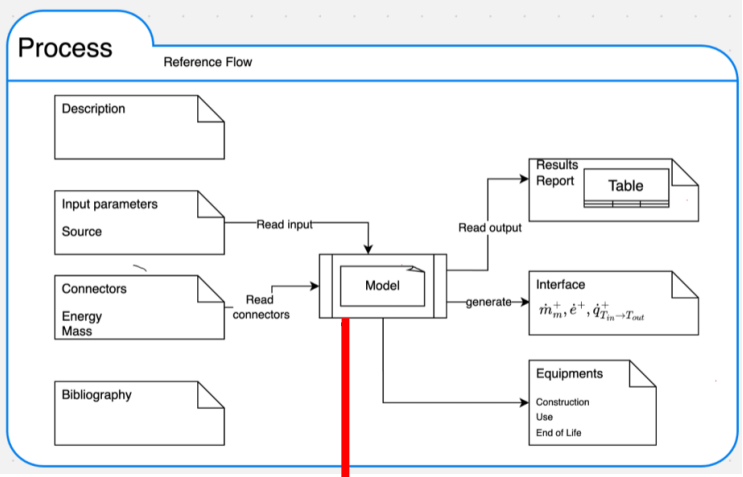
- ✓ Support reproducibility, documentation and maintenance (metadata)
- ✓ Include descriptions of key parameters, variables and assumptions.
- ✓ Accommodate analytical formulas and/or paths to flowsheet variable explorer
- ✓ Retrieve relevant information from units operations, connectors, heat streams and sizes.

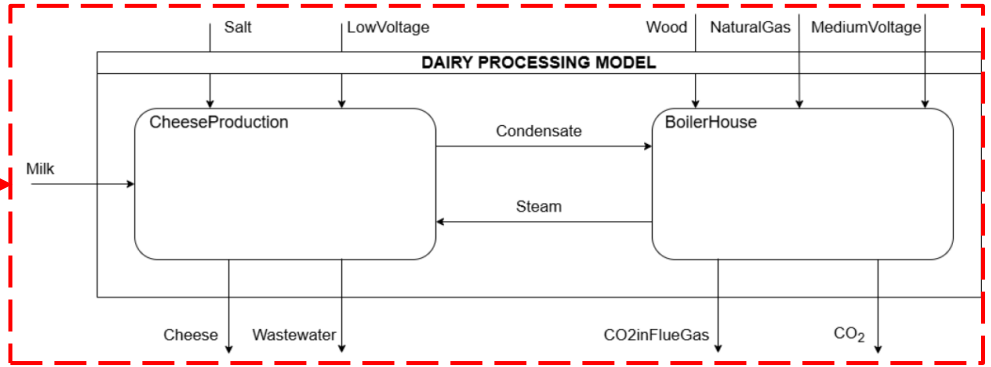


Feedback on the model sharing template

METADATA

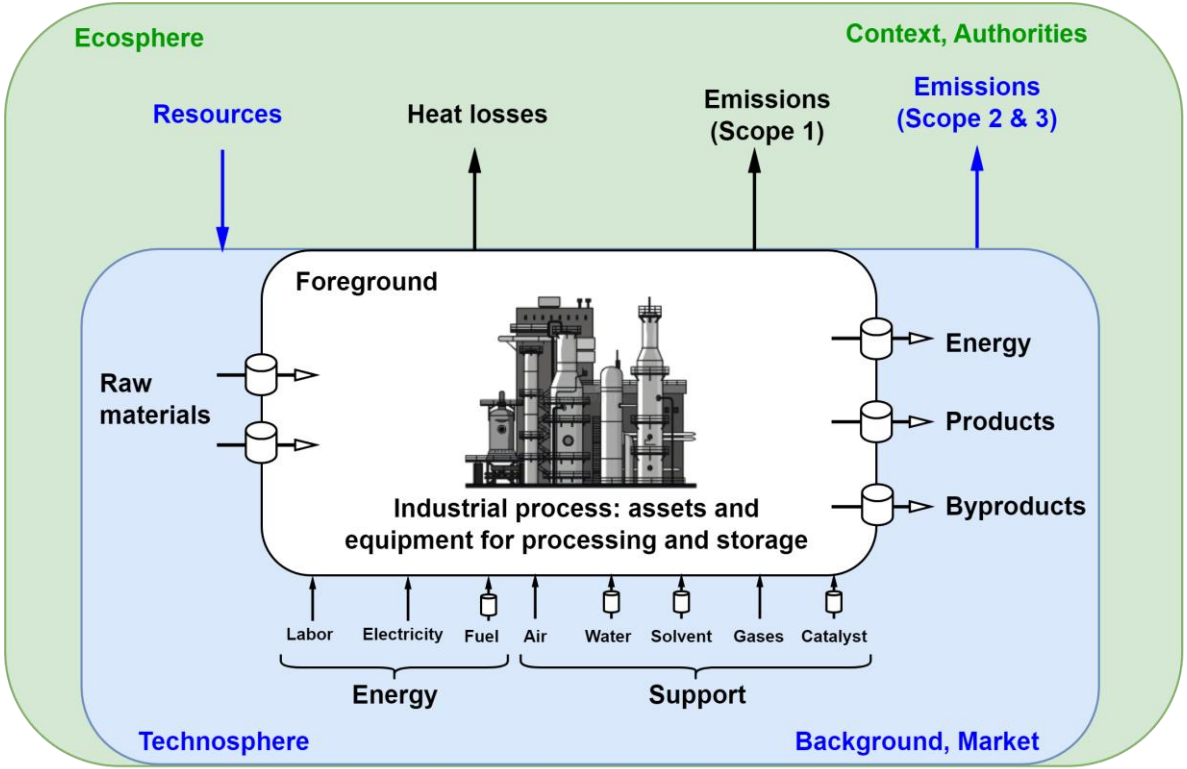
MODEL UID	123-456-789
MODEL NAME	CheeseProcessing
AUTHORS AND CONTRIBUTORS	Daniel Florez-Orrego, Oktay Boztas
CONTACT INFO	daniel.florezorrego@epfl.ch, oktay.boztas@epfl.ch
CREATION DATE	01.06.2025
LAST UPDATED	18.06.2025
MODEL STATUS	Maintained
VERSION	1.0.1
CONFIDENTIALITY	Open
SHARING LAYER	4
GRADE	GRAY-BOX
REFERENCE CAPACITY (MAIN PRODUCT/FEEDSTOCK)	10 kg/h Cheese
TRL	9
KEYWORDS	Cheese, Food
DESCRIPTION	This model represent a dairy process for cheese production from fresh milk, using a wood and natural gas powered boiler house for utility supply. The boiler house unit also includes an amine-based post-combustion carbon capture system.
DOI	https://doi.org/10.0/9887.663A
MAIN RELATED PUBLICATION	https://linktomypublication.com
REFERENCES	cheesefactory.bib
SOFTWARE	ASPEN
SOFTWARE FILE	CheesePlant.bkp
BLOCK FLOW DIAGRAM (.png or .svg)	local/path/to/CheeseProcessing.svg
URL	www.gitlab.epfl.ch/cheeseprocessing
SUPPLEMENTARY MATERIAL	See SUPPL MATERIAL tab





Metadata for univocally defining version, authorship, confidentiality, grade of detail, reference capacity, and software used.

Feedback on the model sharing template

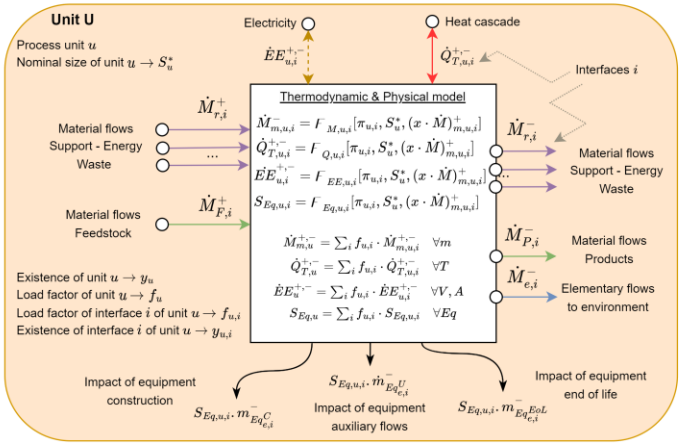


- Industrial process in a technosphere and an ecosphere.
- Background characteristics defines the selection of operating conditions in the foreground.
- Processes operate within a regulatory context of authorities and compliance requirements.

Model interfaces definition

- **Metadata**: Identifier, authorship, versioning, confidentiality.
- **Context**: Connections, flows, quality, cost, impact, TRL.
- **Models**: Parameters, decision variables and assumptions.
- **Metrics**: Energy, environmental, economic indicators.

Black
Grey
White



System integration

$$\min_{\pi^D_{u,i}, f_u, f_{u,i}, y_u, y_{u,i}} KPI_o = \left(\sum_{m,u,i} f_{u,i} \dot{M}^{+,-}_{m,u,i} \cdot v_{m,o} + \sum_{u,i} f_{u,i} \dot{E}E^{+,-}_{u,i} \cdot v_{EE,o}\right) \cdot t_{op} + \sum_{Eq,u,i} f_{u,i} S_{Eq,u,i} \cdot v_{Eq,o}$$

Mass balance over all units u: $\sum_{m,u,i} f_{u,i} \dot{M}^{+,-}_{m,u,i} = 0$

Electricity balance over all units u: $\sum_{u,i} f_{u,i} \dot{E}E^{+,-}_{u,i} = 0$

Heat cascade over all units u: $\sum_{u,i} f_{u,i} \dot{Q}^{+,-}_{T,u,i} + \dot{R}^{+}_{T} - \dot{R}^{-}_{T-1} = 0, \quad \dot{R}^{+}_{T} \geq 0 \quad \forall T$

Load factors upper and lower bounds: $f_{min,u} \cdot y_u \leq f_u \leq y_u \cdot f_{max,u}$

Selection of interface for each unit u:

$$f_{min,u} \cdot y_{u,i} \leq f_{u,i} \leq y_{u,i} \cdot f_{max,u}, \quad \sum_i y_{u,i} \leq y_u, \quad f_{u,i} \leq f_u \quad \forall u, i$$

S^*_u : nominal size of unit u
$S_{Eq,u,i}$: size of equipment Eq of interface i of unit u
$y_{u,i}$: existence of interface i of unit u
$f_{u,i}$: load factor of interface i of unit u
y_u : existence of unit u
f_u : load factor of unit u
$\pi_{u,i}$: modeling parameters of interface i of unit u: $\{D$ decision, L literature, M measurement, C context$\}$

$S_{Eq,u}$
Sizes of the equipment of the overall system

$\dot{M}^{+,-}_{m}, \ \dot{E}E^{+,-}$
Mass and electicity of imported/exported resources of overall system

Performance assessment

Operating expenditure

$$\left(\sum_m \dot{M}^{+,-}_{m} \cdot v_{m,c} + \dot{E}E^{+,-} \cdot v_{EE,c}\right) \cdot t_{op}$$

Capital expenditure

$$\sum_{Eq,u} \frac{1}{\tau_{Eq,u}} \cdot I(S_{Eq,u})$$

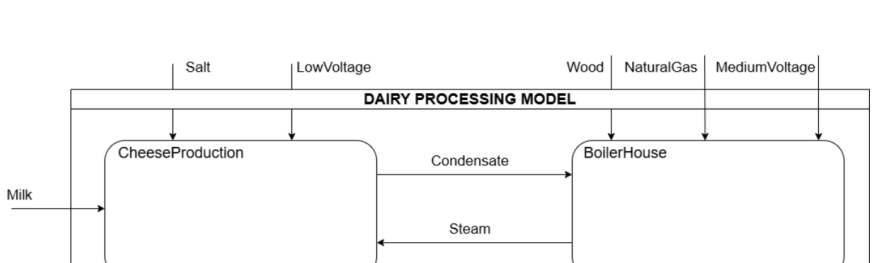
Environmental impact

$$\sum_{Eq,u} S_{Eq,u} \cdot v_{Eq,e} + \left(\sum_m \dot{M}^{+,-}_{m} \cdot v_{m,e} + \dot{E}E^{+,-} \cdot v_{EE,e}\right) \cdot t_{op}$$

Feedback on the model sharing template

CONNECTORS

- External connectors → Interface the specific models with the global database of connectors.
- Environmental and economic impact of flows: It is subject to the study location and users → KPIs coefficients
- Multi-dimensional matrix of KPI coefficients related to each flow for each dimension (thermodynamic, economic, environmental, social, etc.).



EXTERNAL CONNECTORS

UID	NAME (ALIAS)	TYPE	FLOW UNIT	TEMPERATURE	TEMPERATURE UNIT	PRESSURE	PRESSURE UNIT	VAPOR FRACTION [MASS]	PHASE	COMPOSITION [MOLE/MASS]	DESCRIPTION
MILK-FAT3	MILK	MASS	kg/h	25	C	1	bar	-1	(L)	MASSFRAC:FAT[0.03]&H2O[0.97]	Infeed milk flow into the cheese production unit
CHEESE-SOFT	CHEESE	MASS	kg/h	25	C	1	bar	-1	(S)	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	Produced cheese flow from the cheese production unit
SALT-NACL	SALT	MASS	kg/h	25	C	1	bar	-1	(S)	MASSFRAC:NACL[1]	Consumed salt flow into the cheese production unit
WASTEWATER-SEWAGE1	WASTEWATER	MASS	kg/h	25	C	1	bar	-1	(L)	MASSFRAC:H2O[0.99]&FAT[0.01]	Produced waste water flow from the cheese production unit
TAP-WATER	WATER	MASS	kg/h	25	C	1	bar	-1	(L)	MASSFRAC:H2O[1]	Optionally consumed tap water flow into the boiler house
WOOD-DRY	WOOD	MASS	kg/h	25	C	1	bar	-1	(S)	MASSFRAC:C[0.47]&H[0.06]&O[0.45]&S[0]&ASH[0.02]	Consumed wood fuel flow into the boiler house
NATURALGAS	CH4	MASS	kg/h	5	C	1	bar	-1	(V)	MASSFRAC:CH4[1]	Consumed natural gas fuel flow into the boiler house
ELEC-50KV	MEDIUMVOLTAGE	ELECTRICITY	kW	-	-	-	bar	-	-	-	Consumed electricity 50 kV flow into the boiler house
ELEC-13KV	LOWVOLTAGE	ELECTRICITY	kW	-	-	-	bar	-	-	-	Consumed electricity 13 kV flow into the cheese production unit
FLUEGAS-M	CO2INFLUEGAS	EMISSIONS	kg/h	120	C	1	bar	-1	(V)	MOLEFRAC:CO2[0.15]&H2O[0.12]&N2[0.73]	Non-captured CO2 in flue gas flow from the boiler house
CO2-PURE	CO2	EMISSIONS	kg/h	25	C	1	bar	-1	(V)	MASSFRAC:CO2[1]	Captured pure CO2 flow from the boiler house

INTERNAL CONNECTORS

UID	NAME (ALIAS)	TYPE	FLOW UNIT	TEMPERATURE	TEMPERATURE UNIT	PRESSURE	PRESSURE UNIT	VAPOR FRACTION [MASS]	PHASE	COMPOSITION [MOLE/MASS]	DESCRIPTION
STEAM-3BAR-CHEESE	STEAM	MASS	kg/h	SAT	C	3	bar	1	(V+L)	MASSFRAC:H2O[1]	Steam vapor flow into the cheese production unit
COND-3BAR-CHEESE	CONDENSATE	MASS	kg/h	SAT	C	3	bar	0	(V+L)	MASSFRAC:H2O[1]	Condensate flow from the cheese production unit

Feedback on the model sharing template

Energy Audit

Motivation

- Energy role/drivers
- Carbon role/drivers
- Monitoring
- Losses

Innovation

- Smart operation
- Novel resources
- Novel processes
- Novel technologies

Efficiency

Efficiency

- Mass and heat recovery

Waste heat valorization

- Heat pumps
- Waste heat to power or cold

Conversion

Energy vectors

- Renewable resources
- Combined heat and power
- CO_2 capture, use and sequestration
- Industrial sectors
- Chemical and fuels

Systems integration

Multi-energy systems

- Mass and energy grids
- Storage: Power to tank
- Flexibility services to grid
- Multi-actors

Systemic integration

- District heating
- Industrial symbiosis
- Waste management
- Circular economy

Metrics

Thermodynamics

- Energy efficiency
- Mass efficiency
- Exergy destruction

Economics

- Capex - Opex
- TRL
- ROI, Payback

Environmental/Social

- Carbon footprint
- Renewability
- Circularity
- Well-being











Raw materials

Raw materials

Biomass

Water

Hydrogen

Electricity

Gas

CO_2

Energy infrastructure

Products and byproducts

District heating



End of life

Waste management and recycling



Water

Sequestration



Feedback on the model sharing template

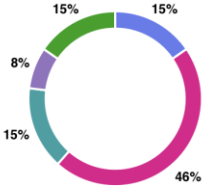
VARIABLES

User grade: It is defined to guide the users of the models to configure and adapt the models

Temporal variations in a model ***(multiperiod, multitime approach)***

26. How would you handle models data sharing when they include batch, partial load, or dynamic aspects:

No applicable (use of steady state models only)	2
Provide typical days / temporal profiles	6
Provide time averaged values	2
Provide values of few representative time periods (snapshots)	1
Other	2



Engineering variables must be physically well-defined, assumed or calculated (software, formula, etc.), as well as referenced.

VARIABLES

VARIABLE NAME	TYPE	FULL NAME	USER GRADE	DEFAULT VALUE OR FORMULA OR PATH TO SC	PHYSICAL UNIT	MIN BOUND	MAX BOUND	LATEX SYNTAX	REFERENCE	DESCRIPTION
MILK_flow	MEASURED	Milk inlet flowrate	MANDATORY	100.00	kg/h	50.00	150.00	F^{in}_{milk}	@Florez-Orrego et al	The amount of fresh milk delivered to the dairy plant
SALT_flow	MEASURED	Salt inlet flowrate	MANDATORY	1.00	kg/h	0.50	2.00	F^{in}_{salt}	@Boztas et al	The amount of salt required for the production of cheese
CHEESE_flow	MEASURED	Cheese outlet flowrate	MANDATORY	10.00	kg/h	5.00	40.00	F^{out}_{cheese}	@Ribeiro Domingos et al	The amount of cheese that is produced from the milk
WASTEWATER_flow	MEASURED	Waste water outlet flowrate	MANDATORY	91.00	kg/h	50.00	100.00	F^{out}_{WW}	@Bhatnagar and Baibhav	s further treatment before being disposed into the environment
LOWVOLTAGE_flow	MEASURED	Low voltage electricity requirement	MANDATORY	3.00	kW	1.50	4.50	E^{in}_{elecLV}	@Florez-Orrego et al	ie low voltage electricity required in the dairy plant for pumping
STEAM_flow	MEASURED	Steam consumption flowrate	MANDATORY	74.69	kg/h	37.35	112.04	F^{in}_{steam}	@Boztas et al	Steam consumption of the dairy plant
CONDENSATE_flow	MEASURED	Return condensate flowrate	MANDATORY	74.69	kg/h	37.35	112.04	F^{out}_{cond}	@Ribeiro Domingos et al	The condensate return going back to the boiler house
pasteurization_heating_Tin	MEASURED	Milk inlet temperature	MANDATORY	25 (or >>/Streams/S27/TEMP)	°C	12.50	37.50	$T^{in}_{milk,past}$	@Florez-Orrego et al	The inlet temperature of the milk to be pasteurized
pasteurization_heating_Tout	MEASURED	Milk outlet temperature	MANDATORY	70.00	°C	35.00	105.00	$T^{out}_{milk,past}$	@Boztas et al	The target temperature for the pasteurization of the milk
pasteurization_heating_Hin	MEASURED	Milk inlet enthalpy flow	MANDATORY	0.00	kW	0.00	0.00	$H^{in}_{milk,past}$	@Ribeiro Domingos et al	Enthalpy content of the milk at the inlet temperature
pasteurization_heating_Hout	MEASURED	Milk outlet enthalpy flow	MANDATORY	5 (or <</Block/HEX101/QCALC)	kW	2.50	7.50	$H^{out}_{milk,past}$	@Bhatnagar and Baibhav	Enthalpy content of the milk at the outlet temperature
pasteurization_heating_cp	LITERATURE	Spec heat capacity of milk to pasteurization	OPTIONAL	4.00	kJ/kgK	2.00	6.00	$c^{milk,past}_{p}$	@Florez-Orrego et al	Specific heat capacity of the milk
pasteurization_heating_ms	MEASURED	Mass flow of milk to pasteurization	MANDATORY	100.00	kg/h	50.00	150.00	$m^{heat}_{milk,past}$	@Boztas et al	Amount of milk to be pasteurized
pasteurization_heating_dTmin	LITERATURE	Milk dTmin contribution	OPTIONAL	5.00	°C	2.50	7.50	$\Delta T_{\min}^{milk}$	@Ribeiro Domingos et al	ifference contribution to the heat exchange by the milk stream
pasteurization_heating_alpha	LITERATURE	Milk heat transfer coefficient	PRIVATE	0.50	kW/m2K	0.25	0.75	α^{milk}	@Florez-Orrego et al	oefficient contribution to the heat exchange by the milk stream
pasteurization_quenching_Tin	MEASURED	Milk quenching inlet temperature	MANDATORY	70.00	°C	35.00	105.00	$T^{in}_{milk,quench}$	@Florez-Orrego et al	et temperature of the milk to be quenched after pasteurization
pasteurization_quenching_Tout	MEASURED	Milk quenching outlet temperature	MANDATORY	25.00	°C	12.50	37.50	$T^{out}_{milk,quench}$	@Boztas et al	et temperature of the milk to be quenched after pasteurization
pasteurization_quenching_Hin	MEASURED	Milk quenching inlet enthalpy flow	MANDATORY	5.00	kW	2.50	7.50	$H^{in}_{milk,quench}$	@Ribeiro Domingos et al	Enthalpy content of the milk at the inlet temperature
pasteurization_quenching_Hout	MEASURED	Milk quenching outlet enthalpy flow	MANDATORY	0.00	kW	0.00	0.00	$H^{out}_{milk,quench}$	@Bhatnagar and Baibhav	Enthalpy content of the milk at the outlet temperature
pasteurization_quenching_cp	MEASURED	Spec heat capacity of milk to quenching	OPTIONAL	4.00	kJ/kgK	2.00	6.00	$c^{milk,quench}_{p}$	@Florez-Orrego et al	Specific heat capacity of the pasteurized milk
pasteurization_quenching_ms	MEASURED	Mass flow of milk to quenching	MANDATORY	100.00	kg/h	50.00	150.00	$m^{cool}_{milk,quench}$	@Boztas et al	Amount of milk to be quenched after pasteurization
pasteurization_quenching_dTmin	LITERATURE	Milk dTmin contribution	OPTIONAL	5.00	°C	2.50	7.50	$\Delta T_{\min}^{milk}$	@Ribeiro Domingos et al	ence contribution to the heat exchange by the pasteurized milk
pasteurization_quenching_alpha	LITERATURE	Milk heat transfer coefficient	PRIVATE	0.50	kW/m2K	0.25	0.75	α^{milk}	@Florez-Orrego et al	cient contribution to the heat exchange by the pasteurized milk
evaporation_preheating_Tin	MEASURED	Inlet temperature of milk for evaporation	MANDATORY	25.00	°C	12.50	37.50	$T^{in}_{milk,preh}$	@Florez-Orrego et al	let temperature of the pasteurized milk for the sensible heating
evaporation_preheating_Tout	LITERATURE	Boiling temperature of milk	MANDATORY	90.00	°C	45.00	135.00	$T^{out}_{milk,preh}$	@Boztas et al	get temperature of the pasteurized milk for the sensible heating
evaporation_preheating_Hin	MEASURED	Inlet enthalpy flow of milk before preheating	MANDATORY	0.00	kW	0.00	0.00	$H^{in}_{milk,preh}$	@Ribeiro Domingos et al	thalpy content of the pasteurized milk at the inlet temperature
evaporation_preheating_Hout	MEASURED	Outlet enthalpy flow of milk after preheating	MANDATORY	6.77	kW	3.39	10.16	$H^{out}_{milk,preh}$	@Bhatnagar and Baibhav	halpy content of the pasteuried milk at the outlet temperature
evaporation_preheating_cp	LITERATURE	Spec heat capacity of subcooled water	OPTIONAL	4.12	kJ/kgK	2.06	6.18	c^{H2O}_{p}	@Florez-Orrego et al	Specific heat capacity of the pasteurized milk
evaporation_preheating_ms	MEASURED	Mass flow of subcooled water	MANDATORY	91.00	kg/h	45.50	136.50	$m^{H2O}_{preheat}$	@Boztas et al	Amount of milk to be sensibly heated before phase change
evaporation_preheating_dTmin	LITERATURE	Milk dTmin contribution	OPTIONAL	5.00	°C	2.50	7.50	$\Delta T_{\min}^{milk}$	@Ribeiro Domingos et al	ence contribution to the heat exchange by the pasteurized milk
evaporation_preheating_alpha	LITERATURE	Milk heat transfer coefficient	PRIVATE	0.50	kW/m2K	0.25	0.75	α^{milk}	@Florez-Orrego et al	cient contribution to the heat exchange by the pasteurized milk

Feedback on the model sharing template

UNIT DESCRIPTION
This unit models the production of cheese, at a production capacity of 10 kg/h of cheese, with energy supply by steam at 3 bar.

Reference sizes and upper/lower caps are important to establish the implementation range of the model.

UNIT NAME	MINIMUM CAPACITY MULT	MAXIMUM CAPACITY MULT
CheeseProduction	1	1

CONNECTORS

NAME (ALIAS)	TYPE	DIRECTION	FLOW VALUE	FLOW UNIT	TEMPERATURE VALUE	TEMPERATURE UNIT	PRESSURE VALUE	PRESSURE UNIT	VAPOR FRACTION [MASS]	PHASE	COMPOSITION [MOLE/MASS]	REFERENCE STREAM	DESCRIPTION
MILK	MASS	IN	100.00	kg/h	25	C	1	BAR	-1	(L)	MASSFRAC:FAT[0.03]&H2O[0.97]	Yes	Infeed milk flow into the cheese production unit
SALT	MASS	IN	1.00	kg/h	25	C	1	BAR	-1	(S)	MASSFRAC:NACL[1]	No	Consumed salt flow into the cheese production unit
CHEESE	MASS	OUT	10.00	kg/h	25	C	1	BAR	-1	(S)	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	No	Produced cheese flow from the cheese production unit
WASTEWATER	MASS	OUT	91.00	kg/h	25	C	1	BAR	-1	(L)	MASSFRAC:H2O[0.99]&FAT[0.01]	No	Produced waste water flow from the cheese production unit
LOWVOLTAGE	ELECTRICITY	IN	3.00	kW	-	-	-	-	-	-	-	No	Consumed electricity 13 kV flow into the cheese production unit
STEAM	MASS	IN	74.69	kg/h	SAT	C	3	BAR	1	(V+L)	MASSFRAC:H2O[1]	No	Steam vapor flow into the cheese production unit
CONDENSATE	MASS	OUT	74.69	kg/h	SAT	C	3	BAR	0	(V+L)	MASSFRAC:H2O[1]	No	Condensate flow from the cheese production unit

Mass flows can go into or out from the sub-unit(s) of the model.

EQUIPMENTS

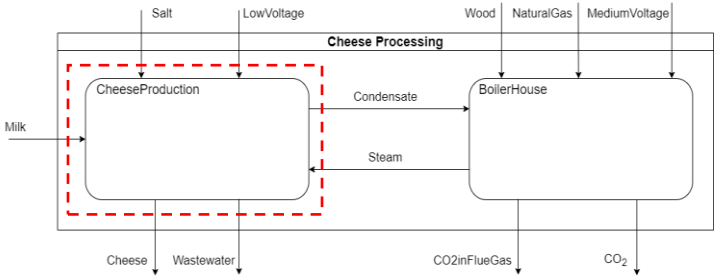
NAME	TYPE	SUBTYPE	REFERENCE	PARAMS	VALUE	UNIT
MilkPump	Pump	Single-stage centrifugal	@Towler and Sinnott	Power	2	kW
				Material	Stainles Steel	-
				Lifetime	20	years
				Date	2025	year
MilkEvaporator	Evaporators	Agitated falling film	@Towler and Sinnott	Area	3000	m2
				Pressure	1	bar
				Material	Stainles Steel	-
				Lifetime	20	years
				Date	2025	year
CheeseProdSys	Asset	-	@Florez-Orrego et al	Capital	75000	EUR
				Lifetime	20	years
				Date	2025	year

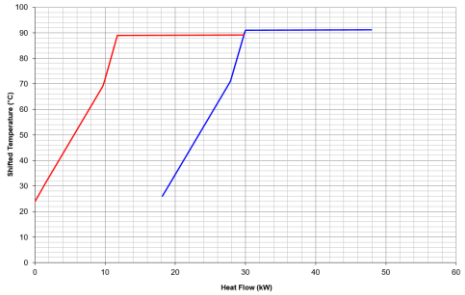
Added CAPEX references, equipment subtype type, and material of construction for all units.

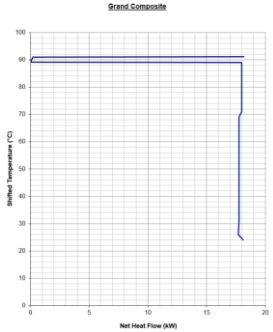
FIXED OPERATING COST [EUR/y]	VARIABLE OPERATING COST [EUR/h]
0	0

Feedback on the model sharing template

Pinch analysis (heat integration) requires the knowledge of the heat cascade and can include different heat cascades (constraints). Allows the characterization of heat streams, waste heat, heat recovery potential and quality (GCC and CC plots).







HEAT STREAMS									
NAME	TYPE	INLET TEMPERATURE VALUE	INLET TEMPERATURE UNIT	OUTLET TEMPERATURE VALUE	OUTLET TEMPERATURE UNIT	INLET ENTHALPY FLOW VALUE	INLET ENTHALPY FLOW UNIT	OUTLET ENTHALPY FLOW VALUE	OUTLET ENTHALPY FLOW UNIT
pasteurization_heating	Cold	25.0	C	70.0	C	0.00	kW	5.00	kW
pasteurization_quenching	Hot	70.0	C	25.0	C	5.00	kW	0.00	kW
evaporation_preheating	Cold	25.0	C	90.0	C	0.00	kW	6.77	kW
evaporation_latentheat	Cold	90.0	C	90.1	C	0.00	kW	18.00	kW
condensation_latent	Hot	90.1	C	90.0	C	18.00	kW	0.00	kW
condensation_subcooling	Hot	90.0	C	25.0	C	2.00	kW	0.00	kW
wastewater_cooling	Hot	90.0	C	25.0	C	4.77	kW	0.00	kW
steam_condensation	Hot	133.1	C	133.0	C	50.00	kW	0.00	kW
cheese_curation	Cold	15.0	C	25.0	C	0.00	kW	0.07	kW

SPEC HEAT CAP VALUE	SPEC HEAT CAP UNIT	MASS FLOW VALUE	MASS FLOW UNIT	DTMIN CONTR VALUE	DTMIN CONTR UNIT	HEAT TRANSFER COEF VALUE	HEAT TRANSFER COEF UNIT	HEAT CASCADE	REFERENCE STREAM	DESCRIPTION
4.00	kJ/kgK	100.00	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Milk pasteurization heating demand
4.00	kJ/kgK	100.00	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Milk pasteurization quenching demand
4.12	kJ/kgK	91.00	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Milk evaporation preheating demand
2410.00	kJ/kgK	26.89	kg/h	2.5	K	1.00	kW/m2K	CHEESEPLANT	No	Evaporation enthalpy of milk concentration
2410.00	kJ/kgK	26.89	kg/h	2.5	K	1.00	kW/m2K	CHEESEPLANT	No	Condensation enthalpy of water from milk concentration
4.12	kJ/kgK	26.89	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Subcooling of water separated in the milk concentration
4.12	kJ/kgK	64.11	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Waste water cooling demand
2410.00	kJ/kgK	74.69	kg/h	2.5	K	1.00	kW/m2K	CHEESEPLANT	No	Heating demand provided by the steam condensation
2.53	kJ/kgK	10.00	kg/h	8	K	0.10	kW/m2K	DEFAULT	No	Cheese curation heating demand (maintain)

Different versions of model sharing template

One general template vs. Special templates

→ Adaptive improvement to different “scales”, simulation approaches, sectors, technologies.

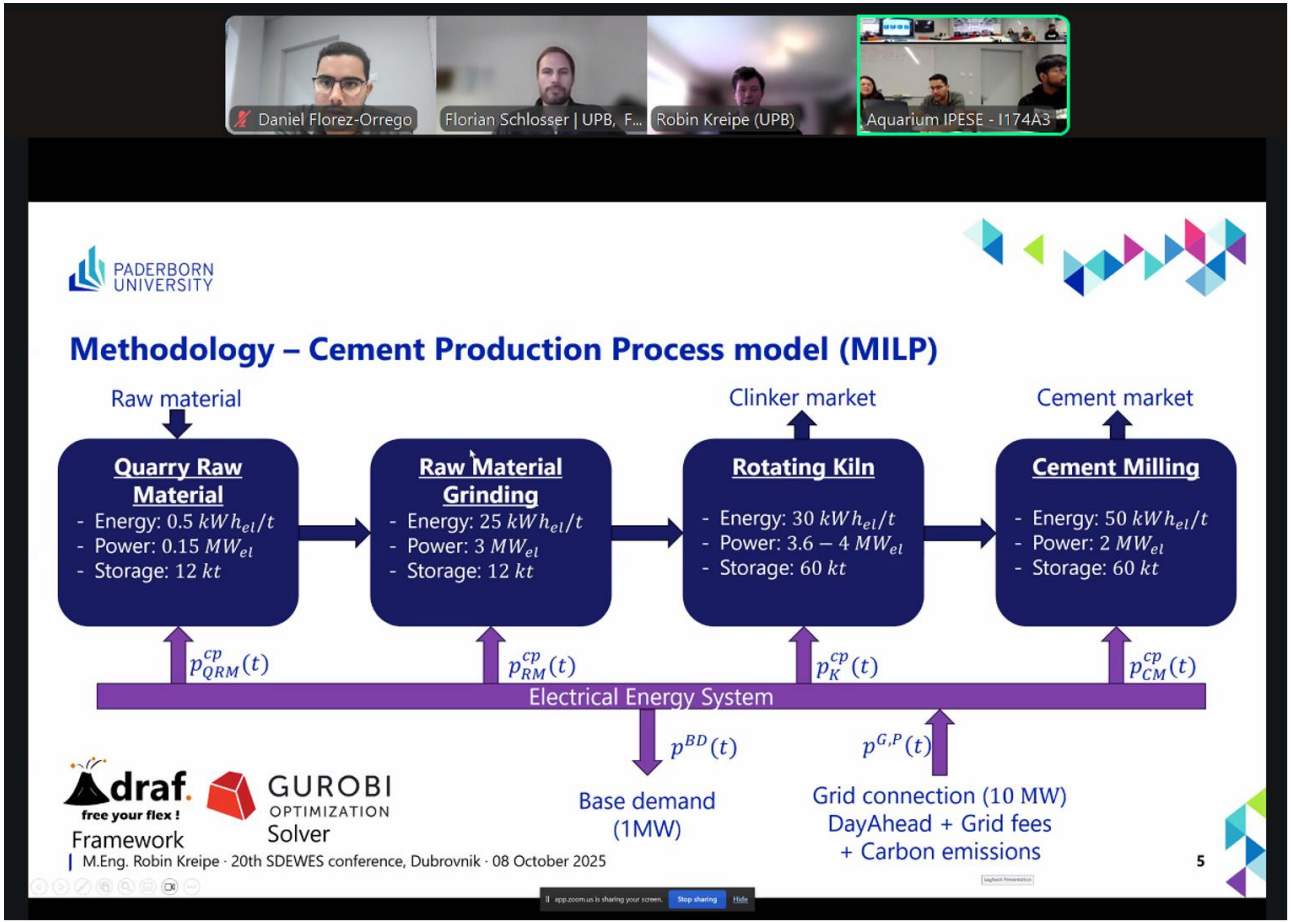
Examples of **specific extensions** in Swiss consortium**:** *Superstructure, Storage, Batch, Solids, Interfaces.*

The following are your choices of “starting point” models. Can you foresee compatibilities or challenges?

Model / Sector	Software	Status
Ammonia plant / Chemicals	Aspen® Plus	Open
Zeotropic mixture heat pump	Python	Design/off-design
Quicklime production	OSMOSE	To be published
Energy storage / reactor modelling	MATLAB® + Python	Published
Electricity market + H_2 production	N/A	Confidential
Dual fluidized bed for plastic waste	Aspen® Plus	Published by June 2026
Agroindustry (Sugar/Ethanol)	Aspen® Plus	Published
Solar energy integration to industrial	TRNSYS®	Unpublished
CO_2 to Methanol / CCU	Aspen® Plus	Unpublished
Cement plant scheduling optimization	Python + Gurobi	Paper Q1 2026

Participants modeling paradigms and frameworks

Germany's MILP modeling framework (Kreipe, Schlosser)



Model interfaces definition

Netherlands' Aspen Models documentation approach (Perez-Fortes, Ramirez)

- An example of an Excel file corresponding to one of our ethylene synthesis models in Aspen Plus (we share the Excel file and a readme file).
- The modelling guidelines we follow in the project (and the overall team as well, in other projects).

README

```
# Process datasheet
**Project:** Unravelling the impacts of using alternative raw materials in industrial clusters
**Creators:**  Tonny Manalal, Michael Tan, Inna Stepchuk, Paola Ibarra-Gonzalez, Mar Pérez-Fortes, Andrea Ramirez Ramirez
**Affiliations:**
- Delft University of Technology, Faculty of Technology, Policy and Management, Department of Engineering, Systems and Services, Section of Energy and Industry
- Delft University of Technology, Faculty of Applied Sciences, Department of Chemical Engineering

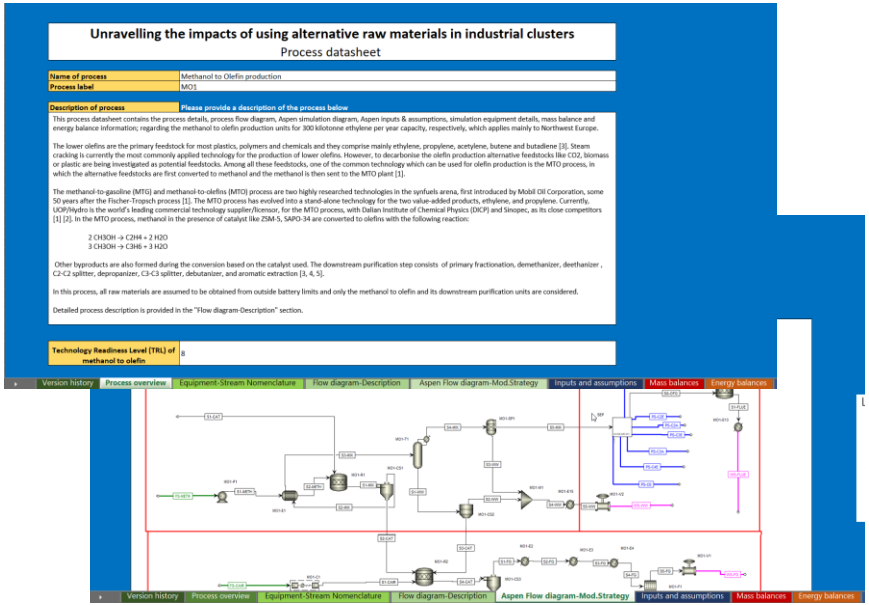
Corresponding creator: Andrea Ramirez Ramirez
Contact Information: c.a.ramirezramirez@tudelft.nl

## Introduction
The process datasheet contains the complete documentation for an Aspen Plus model.
It contains all input data and underlying assumptions, as well as the resulting material and energy balances, and the total bare equipment costs.

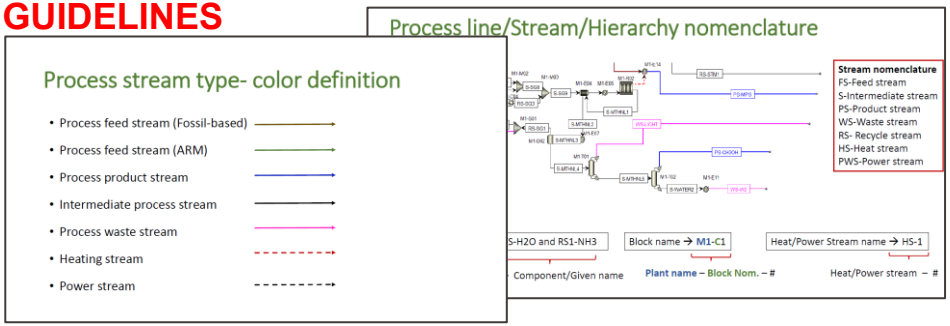
## Data specific information
Below, a description is given for each of the tabs in the Process Datasheets of each process.

### Process overview
Provides an overview of the process. Describes the process technology modeled and the main products of the process.
```

EXCEL



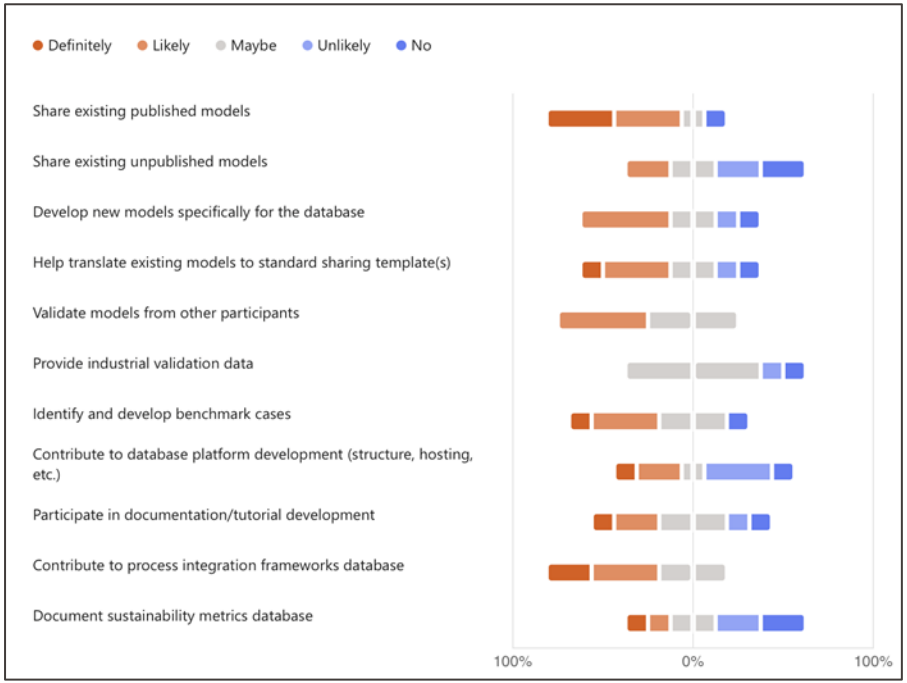
GUIDELINES





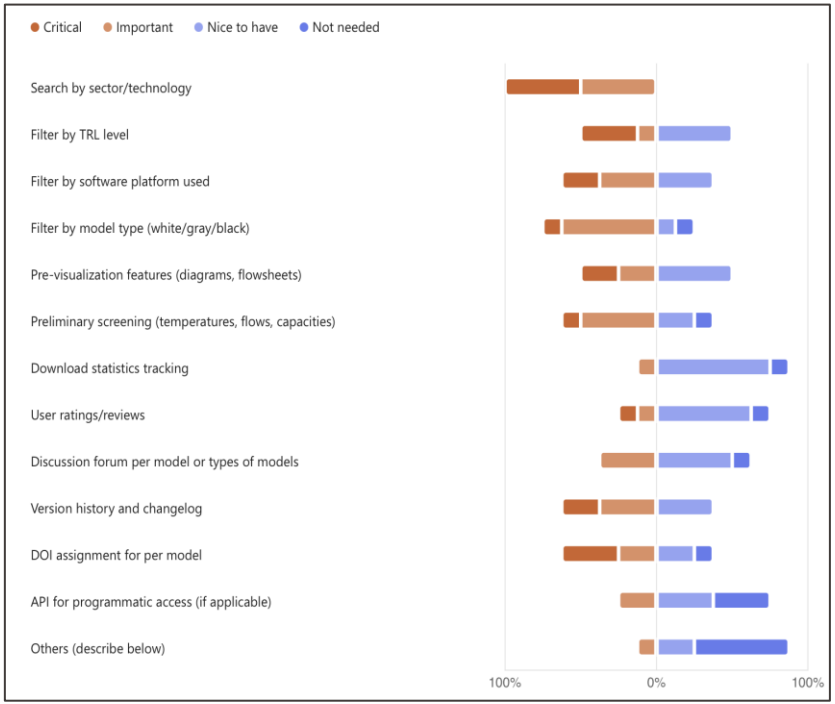
Database infrastructure

Features and infrastructure of database and framework

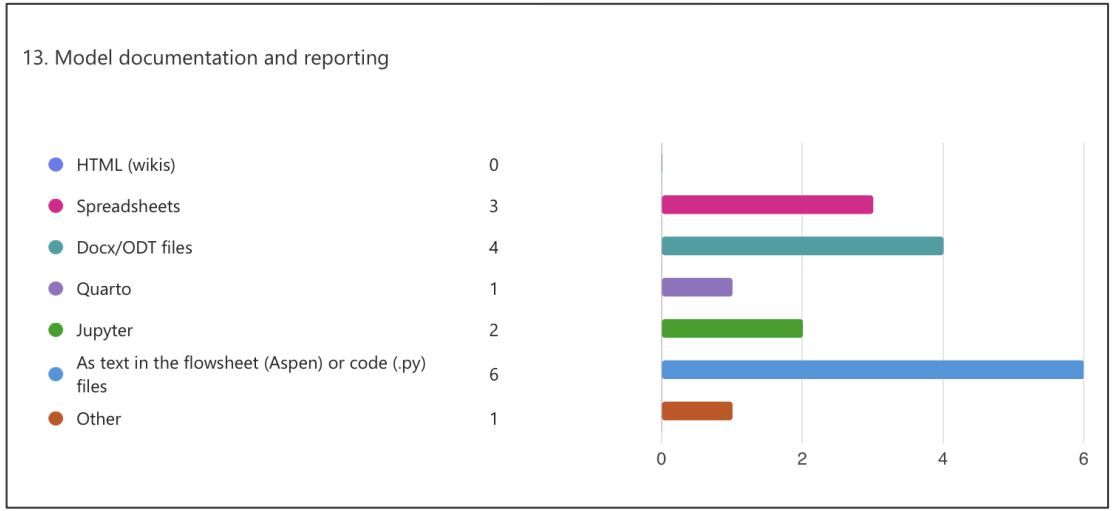


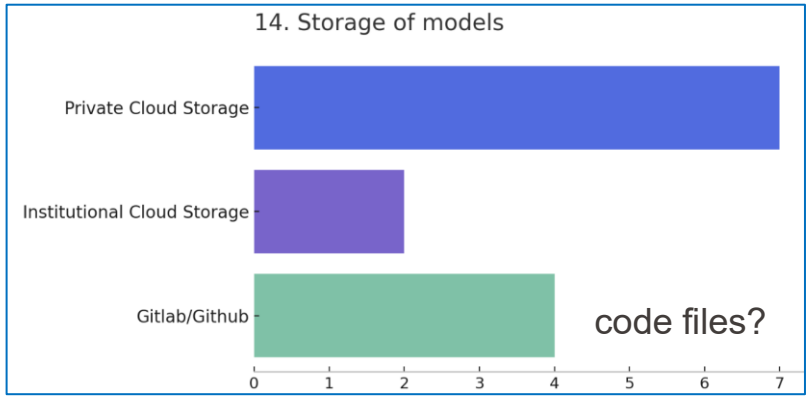
- **Desired features:** search by sector and technology, TRL level, **DOI** of model/database, filter by model type and software.
- **Nice to have feature:** Statistics, review and rating (?) of model and **discussion forum per model** or types of model.

- Excel is used currently as a communication and standardization exercise of model-sharing → not necessarily imposed as the **final solution**



Features and infrastructure of database and framework

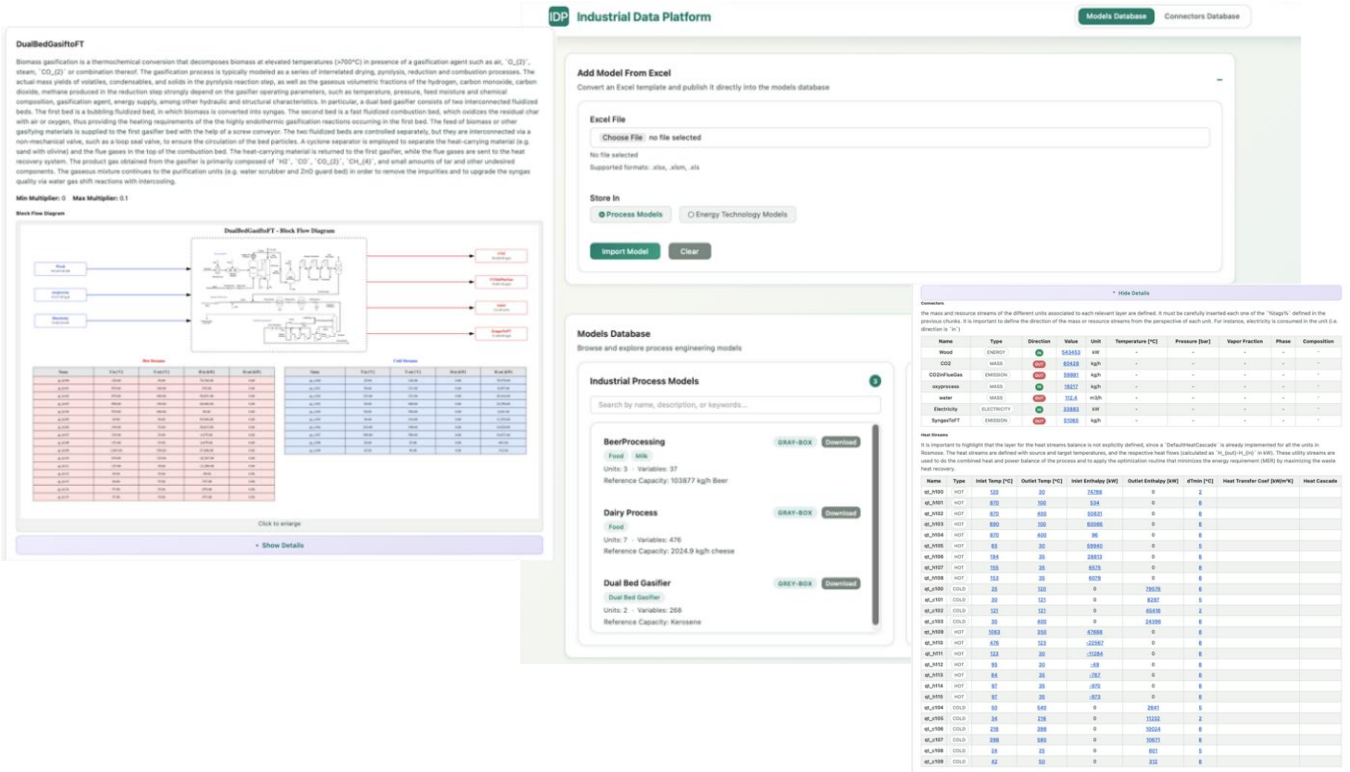




- **Aspen annotation not** easily accessible without licensed software.
- **Submission and storage** must be in an **easy platform** for all the participants.
- A **centralized platform** could standardize model descriptions and host the latest official version → location, cost, space
- Meanwhile **external institutional repositories** can manage detailed development, versioning, and history.

Features and infrastructure of database and framework

- Snapshots of a proposal of submission system of model for database definition:



- Work in progress

Features and infrastructure of database and framework

Syntax, terminology and specifications

Infrastructure, management and maintenance

Data base

Push code

Trigger a pipeline

.gitlab-ci.yml

CI Pipeline

Preparation Build Tests

Deploy

Introduction
Problem Definition
Dairy Process
Cooling Tower
Heat Pump
Visualization of Heat Pump
Heat Pump 2
Heat Pump 3
Heat Pump 4
Market
Optimization Results

Heat Pump

Note: Heat pumping will be profitable only if it allows to transfer heat from below to above the pinch, i.e., if it transforms excess heat from the system heat source into useful heat for the system heat sink.

Layers of the Heat Pump ET

OSMOSE LAYERS heatpump

Layer	Display name	shortname	Unit	Color
ELEC	Electricity	elec	kW	yellow

Heat Pump unit of the Heat pump ET

OSMOSE UNIT heatpump

unit name	type
HeatPump	Utility

Parameters of the Heat Pump unit

cost1	cost2	cinv1	cinv2	imp1	imp2	fmin	fmax
0	0	0	123955.26918427552	0	0	0	10

Heat pump Streams

Resource Streams Defining the resource streams, in this case electricity to the heatpump

layer	direction	value
ELEC	in	656.115107913669

Tin	Tout	Hin	Hout	DT min/2	alpha
55.5	55.5	0	5343.884892086331	2	1
74.5	74.5	6000	0	2	1

ation of Heat Pump

656.115107913669 [kW]

HeatPump

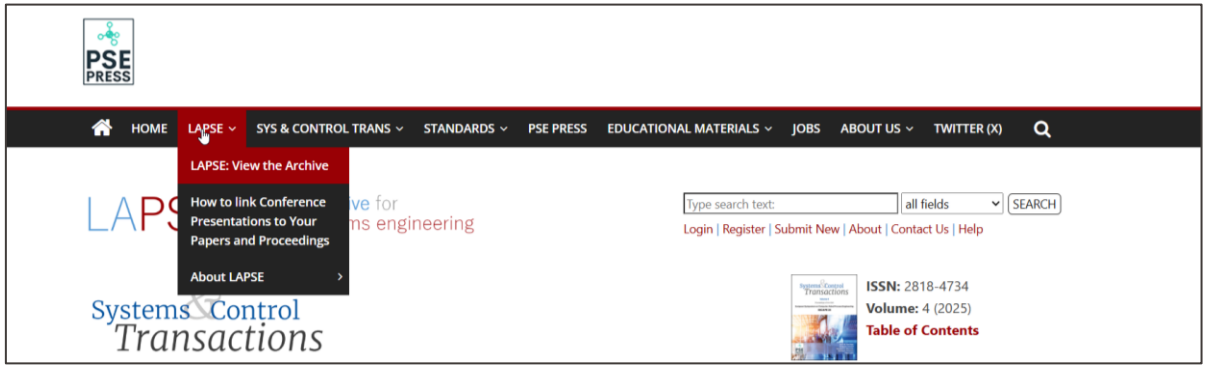
Heat:		
55.5 C	5343.884892086331 kW	55.5 C
74.5 C	6000 kW	74.5 C

Costs:

Op 1:	0
Op 2:	0
Inv 1:	0
Inv 2:	123955.26918427552

heatpump-visualize

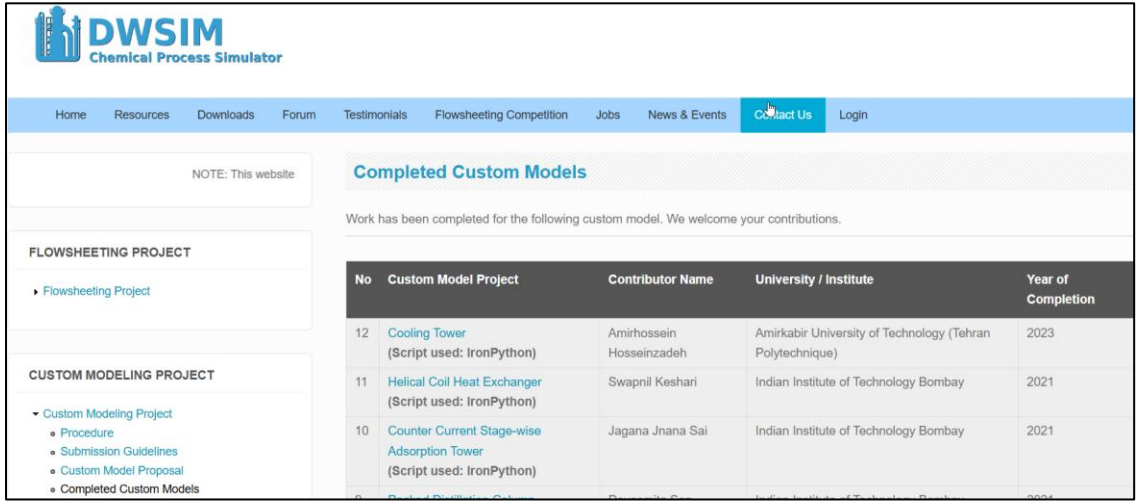
Other model databases: FOSSE and LAPSE



PSE supported initiative (Adams II)

FOSSEE - Free and Open Source Software for Education

(Ramachandran)



Other model databases: MASBOOK

EPFL Master of Advanced Studies: Decarbonizing Industrial Process and Energy Technologies

Preface
1 Tools
2 Thermodynamic properties calculation
3 Brewery audit
4 Dairy audit
5 Distillery audit
6 Explosives production audit
7 Leather tanning audit
8 Malting process
Utility systems models
Decarbonization technology models
Conclusions

Master of Advanced Studies: Decarbonizing Industrial Process and Energy Technologies

Report Summary

AUTHOR
IPESE EPFL

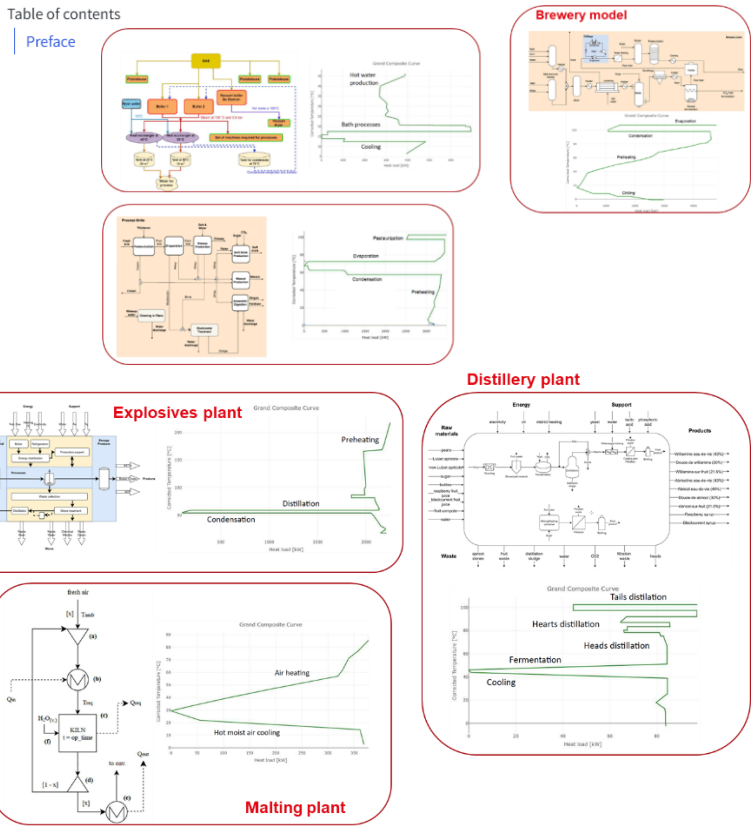
PUBLISHED
July 16, 2025

Preface

> "Our goal is to ensure that all students succeed in this program, learn the theory, and use it in real industrial applications." - Prof. François Marechal

The following contributors worked in the preparation of this book:

- Daniel Alexander Florez-Orrego
- Meire Ellen Gorete Ribeiro Domingos
- Dareen Zuhir Omar Dardor



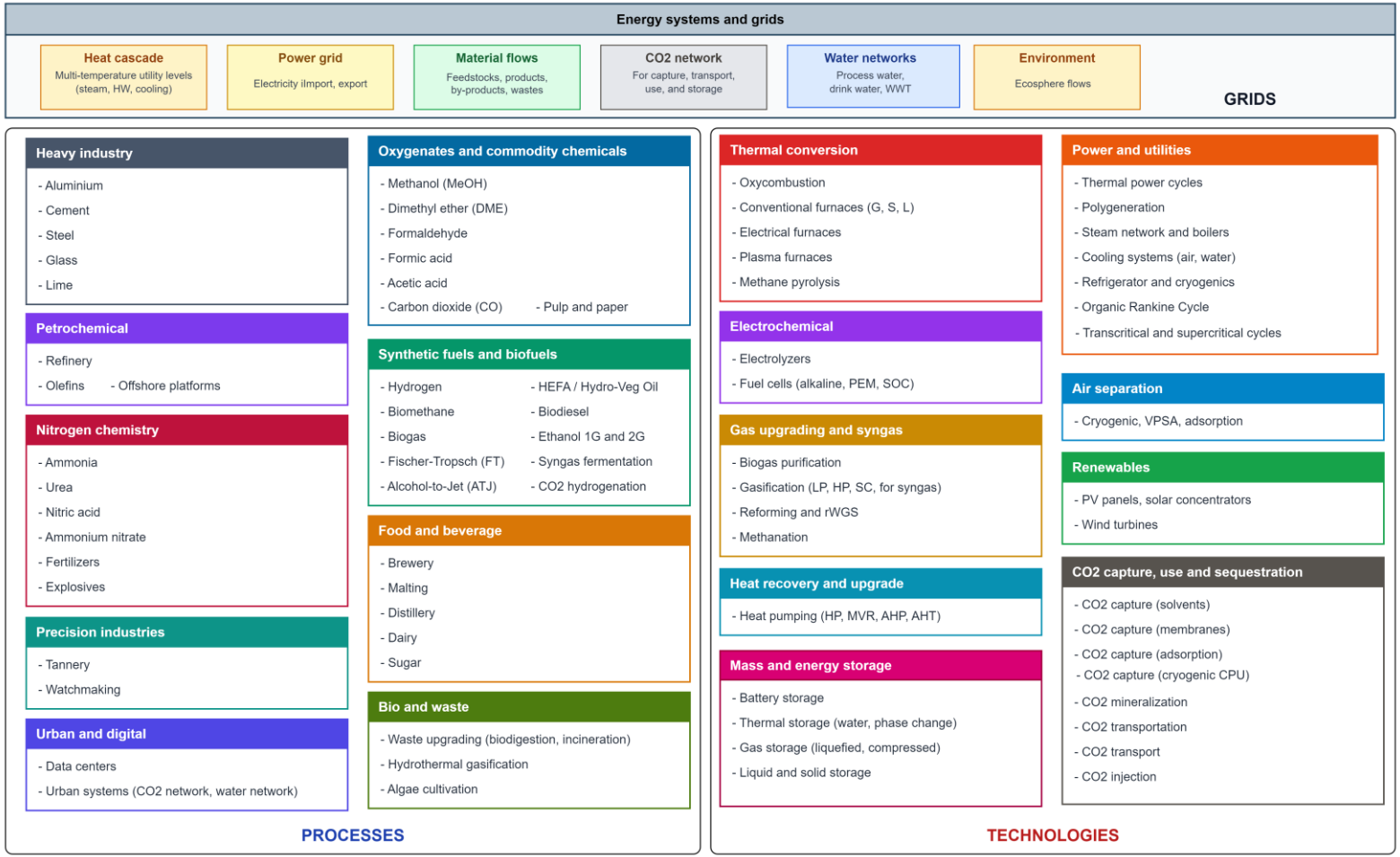
Models of industrial processes and decarbonization technologies

- Dairy
- Brewery
- Explosives
- Malting
- Tannery
- Distillery
- Aluminium
- Sugar
- Olefins
- Watchmaking
- Malting
- Fertilizers
- Cement
- Steel
- Glass
- Refinery
- Methanol
- Dimethyl ether
- Hydrogen
- Biomethane
- Hydro-Veg Oil
- Lime
- Biogas
- Waste upgrading
- Fischer Tropsch
- Alcohol to jet
- Biodiesel
- Ethanol 1&2G
- Hydrothermal gasification
- Data centers
- Urban systems

- Oxycombustion
- Conventional furnaces (G,S,L)
- Electrical furnaces
- Plasma furnaces
- Methane pyrolysis
- Thermal power cycles
  - ICE
  - GT
  - ST
  - ORC
- Polygeneration
- Steam network
- Cooling systems (air, water)
- Air separation (Cryo., VPSA, adsorb.)
- Refrigerator
- Electrolyzers

- Fuel cells (alkaline, PEM, SOC).
- Heat pumping (HP, MVR)
- PV panel
- Wind turbines
- Biogas purification
- CO_2 capture solvents
- CO_2 capture membranes
- CO_2 capture adsorption
- CO_2 mineralization
- CO_2 transportation
- CO_2 injection
- Gasification (syngas)
- Reforming and rWGS
- Methanation

Models of industrial processes and decarbonization technologies





Annual report

Task XXIV Annual Report

- Min. 5 and max. 10 pages (e.g. editable slides, docs or LaTeX formats, although PDF files are also possible):
  - i. brief description of your country's consortium (experience, research focus, other Tasks collaboration, etc.),
  - ii. list of conferences and other events attended, as well as technical and journal publications that are related to the objectives of the Task XXIV,
  - iii. exemplary diagram(s) or snapshots that briefly illustrate your models structure; any raw links to known/used model repositories (own or external) are also relevant,
  - iv. simplified descriptive workflow and illustrative screenshots of your optimization framework(s) (in-house or proprietary software),
  - v. examples of output reports, documentation excerpts, and other relevant content for the Annual Report of the Task XXIV



✓ CONTRIBUTION TO IETS NEWSLETTER (INTERVIEW) → early January

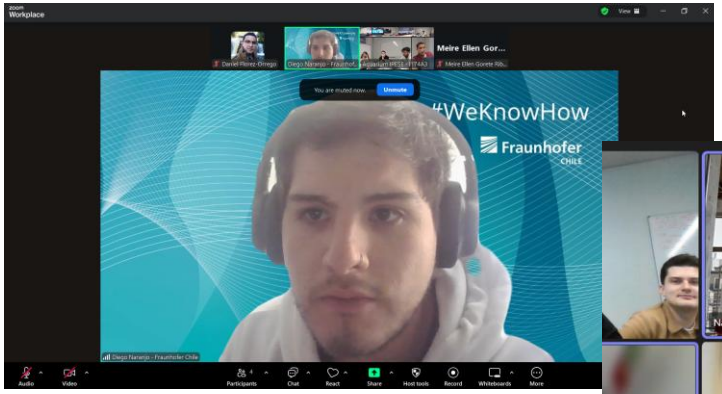
✓ REPORT TO SWISS FEDERAL OFFICE OF ENERGY (SFOE) → mid January

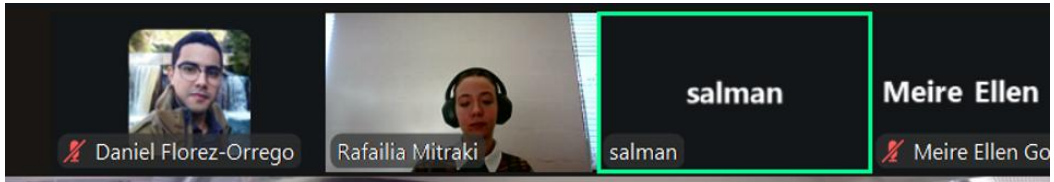
❑ REPORT TO THE IETS SECRETARIAT (DEBRIEFING) → mid February

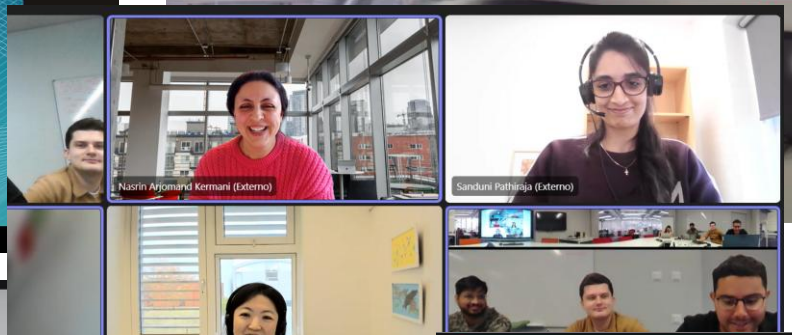


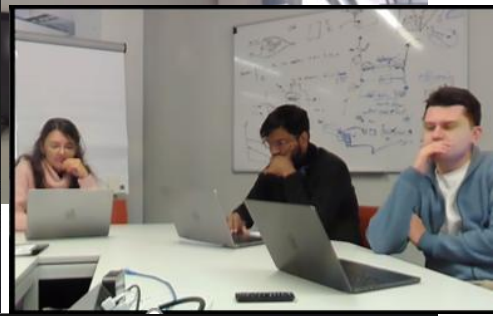
Disemmination

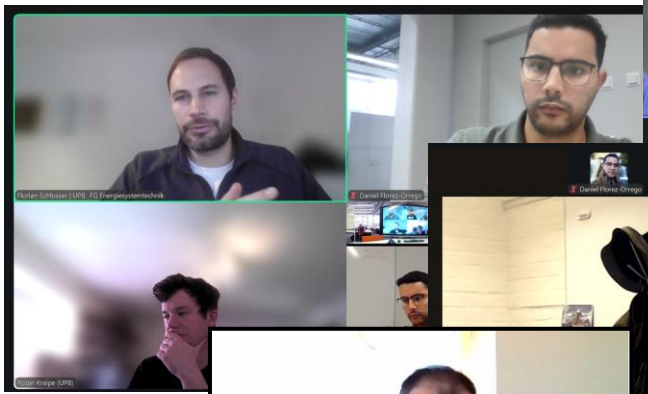
Bilateral meetings



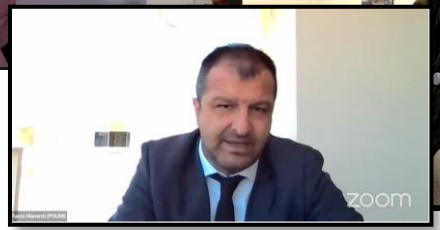








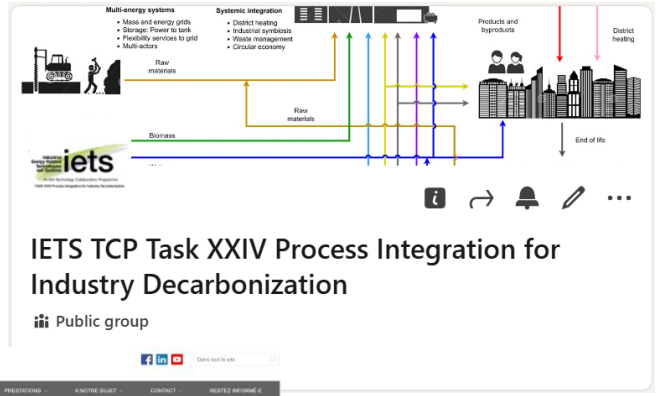


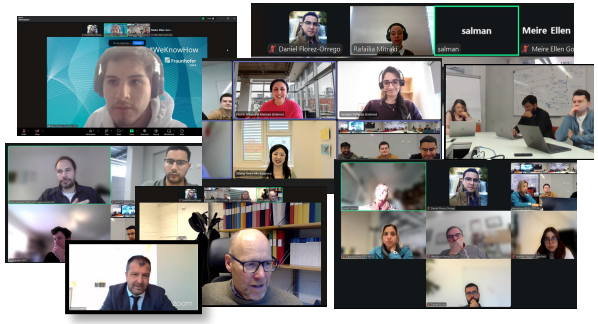


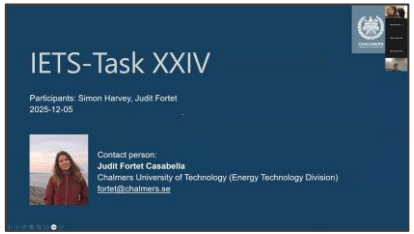


Dissemination

- LinkedIn group https://www.linkedin.com/groups/16699007/









- Intra-task bilateral and plenary meetings, and annual in-person meeting.



- Continuing education programs MAS

- Inter-task Task XI and XXIV workshop on Industrial Symbiosis



- Keynote lecture in ESCAPE 2025, Ghent



- Special session Task XXIV in IBCC 2025, Lisbon







Next steps

Next steps

- Next bilateral meetings on selected model development: advances and discussions
- Database infrastructure: framework scalability and maintainability
- Models validation: sub-teams configuration and peer-reviewing strategy

Hints

- Starting right away. By doing it already, we can see what works and what does not. Complexity is sometimes just another word for activation energy.
- Choose one of your richest possible model with the least amount of sensitive data, and use this exercise e.g. for generating the supplementary data of your next journal paper.
- If main concern is the time, dedicate **2 hours** every **2 weeks**, with discussions meetings every **2 months** (3 bilateral meetings a year + 2 plenary + 1 one annual in-person meeting).
- Associate Task XXIV objectives to your ongoing projects: all your contributions to ongoing projects, theses and workshops count as model sharing and networking activities for the Task.
- Which are your experiences with generating new models? Which models would have been useful to have in hand for your current research?

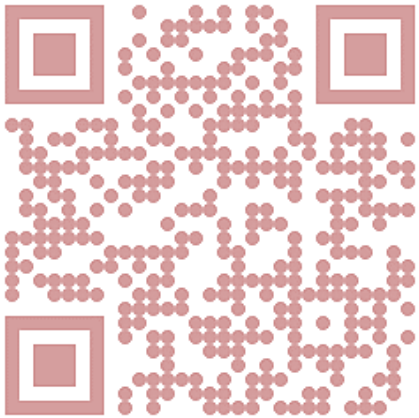


Task XXIV
Process integration for industry decarbonization

Subtask 1 Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics

https://iea-industry.org/tasks/process-integration-for-industry-decarbonization/

Questions

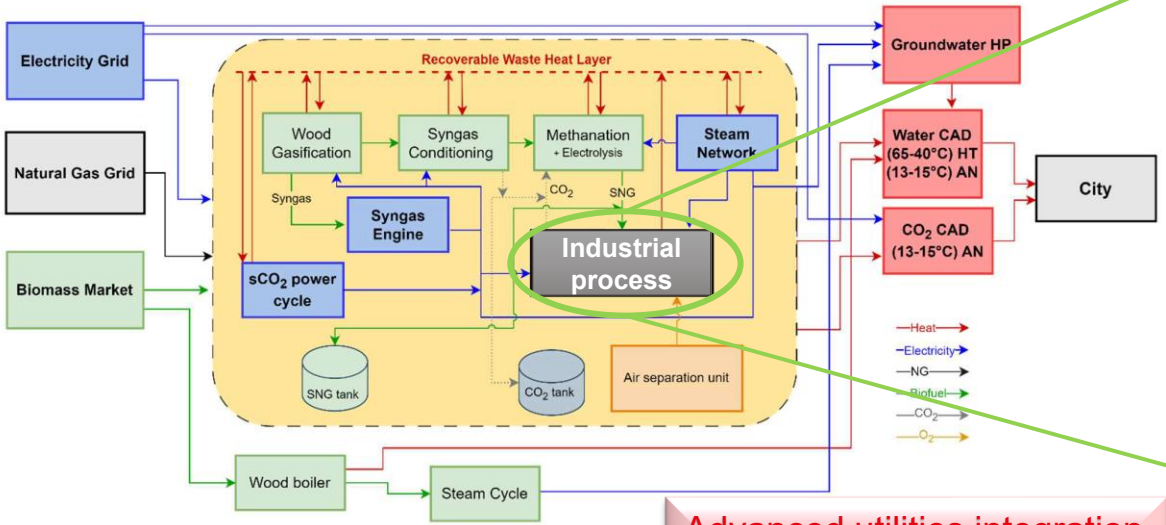


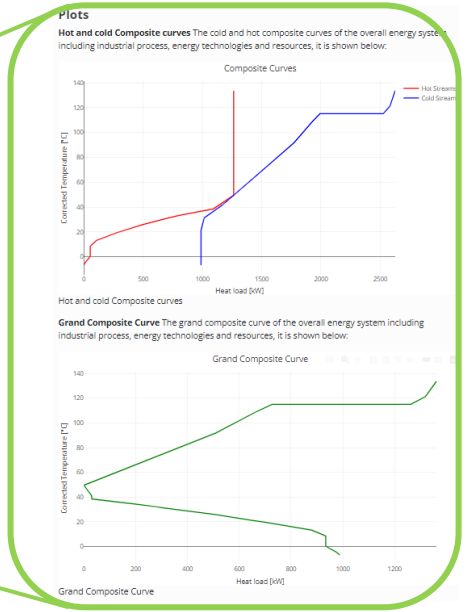
Daniel Florez-Orrego
daniel.florezorrego@epfl.ch



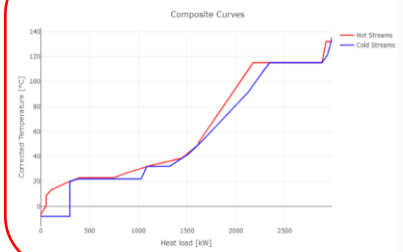
Back up slides

Process Integration and Decarb





Advanced utilities integration



Economic indicators:

Optimized objective: obj_totalcost

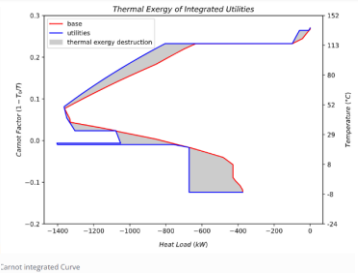
Operational expenditure (CHF/y): 381934

Capital expenditure (CHF/y): 75166

Impact (kg_{CO2}/y): 115946

Total Expenditure (CHF/y): **457099**

Total investment (non-annualized) (CHF): 737987

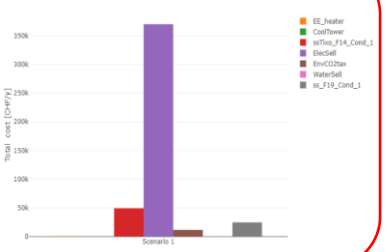


Mass and electricity streams:

The following tables shows the mass and electricity streams, without and with details of the internal streams of the HP superstructure:

Without HP Internals | With HP internals

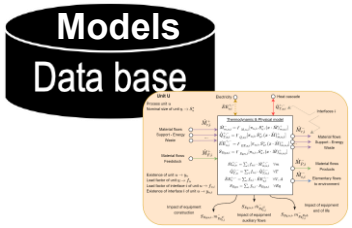
Connection	From	To	Value	Units
Electricity	market_ElecSell	refrigerator_ss_F19_Comp12	56.59	kW
Electricity	market_ElecSell	heatpump_ssTixo_F14_Comp24	88.25	kW
Electricity	market_ElecSell	heatpump_ssTixo_F14_Comp12	231.4	kW
Electricity	market_ElecSell	heatpump_ssTixo_F14_Comp23	30.7	kW
Electricity	market_ElecSell	coolingtower_CoolTower	1.79	kW
Electricity	market_ElecSell	HeaterEE_EE_heater	54.09	kW
EnvCO2Em	market_ElecSell	market_EnvCO2tax	28.99	kg/h
WATER	market_WaterSell	coolingtower_CoolTower	8.29	kg/h



Transparent assumptions, documented calculations, reduced versioning errors, open-source methods and tools.

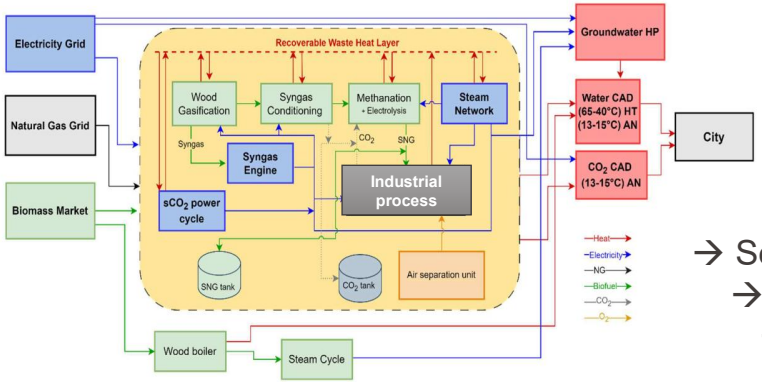
Analyze – Generate – Interpret – Report

Analyze

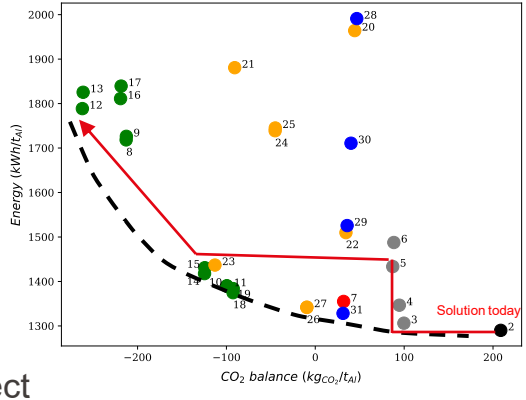


- Innovation
- Technology options
- Ex-ante models
- Interfaces
- Superstructure
- System integration

Generate

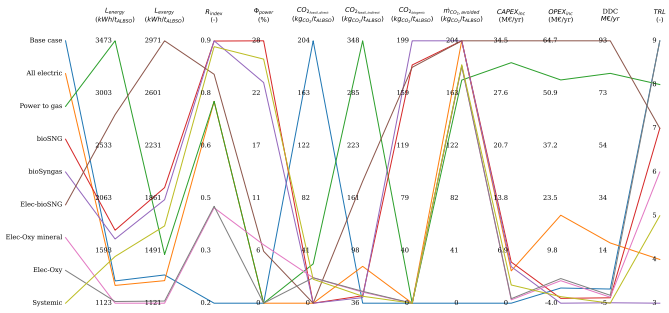


→ Select
→ Size
→ Connect
→ Operate



Interpret

KPIs
- Economic
- Thermodynamic
- Environmental
- Externalities

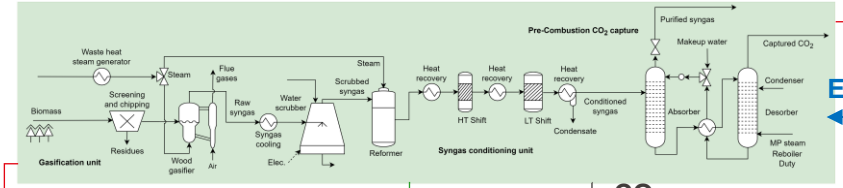


Report

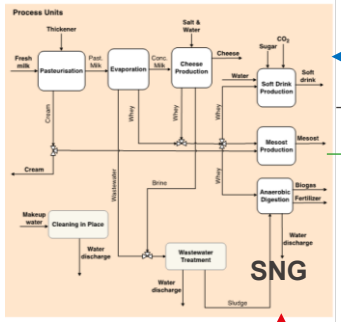
- Multi-criteria
- Trade-offs
- Assess risk
- SDGs
- Planetary limits
- Decision-making

Symbiosis

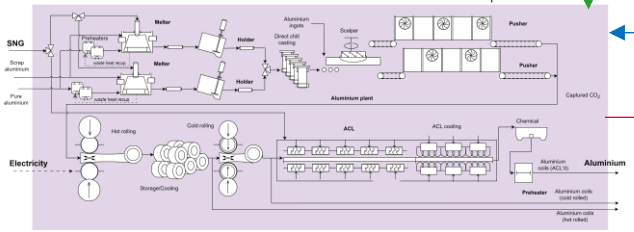
Biomass conversion



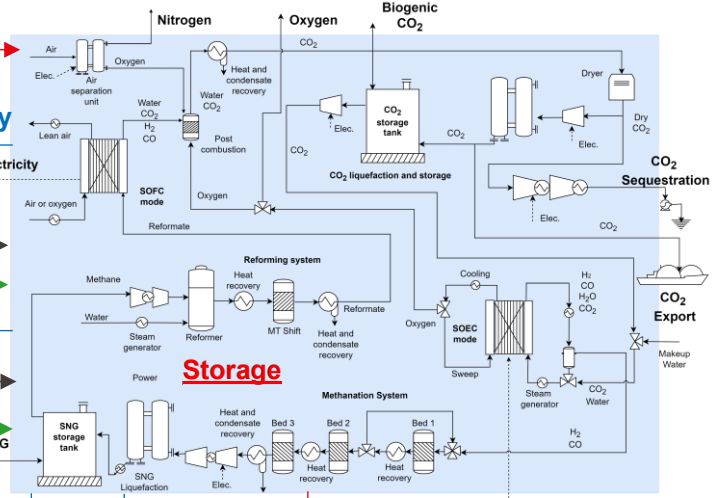
Low T industry



High T industry

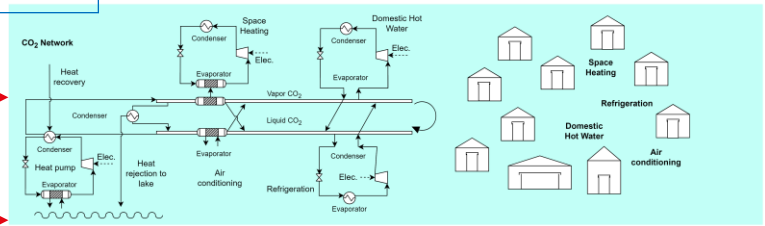


Storage



PtG and Grids

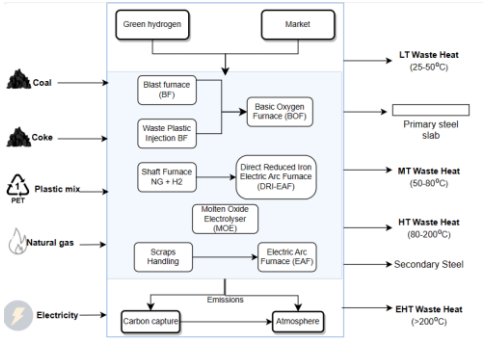
District heating network



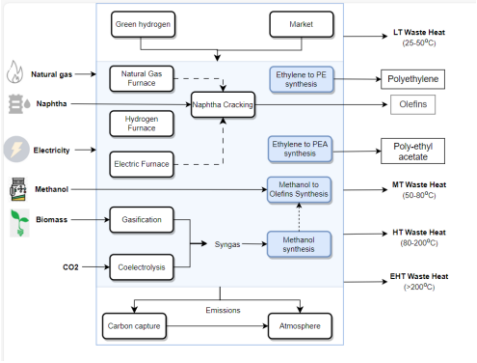
Heat
Electricity
CO_2
Syngas
Electricity
CO_2
SNG
SNG
SNG
CO_2
Heat
Electricity
Heat
Electricity
Heat
Heat

AIDRES Database

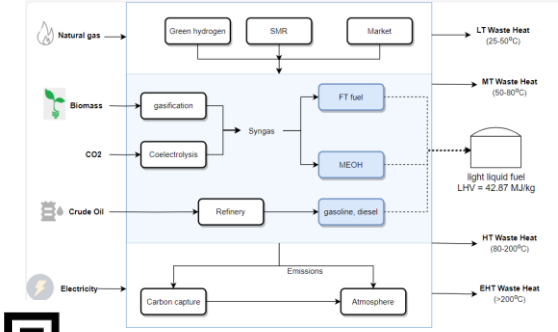
Steel sector



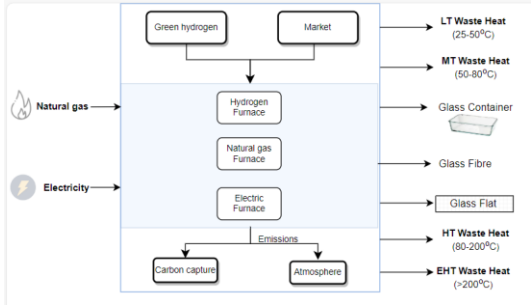
Chemical sector



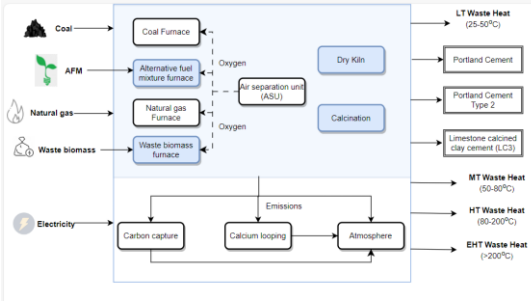
Refineries sector



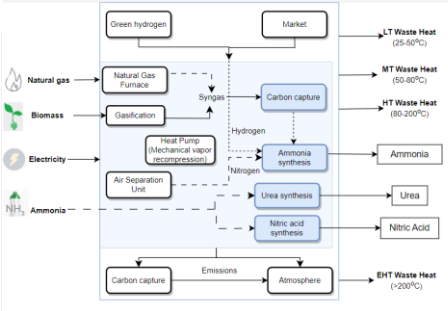
Glass sector



Cement sector



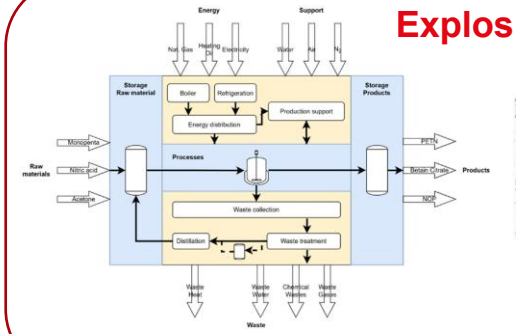
Fertilizer sector

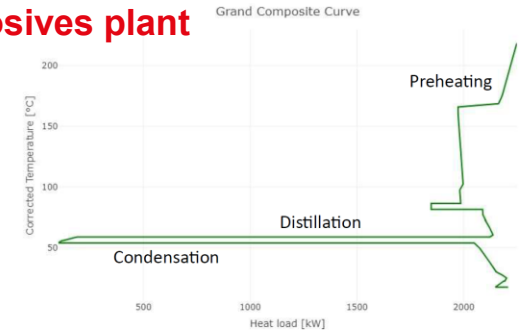


https://op.europa.eu/s/z69G

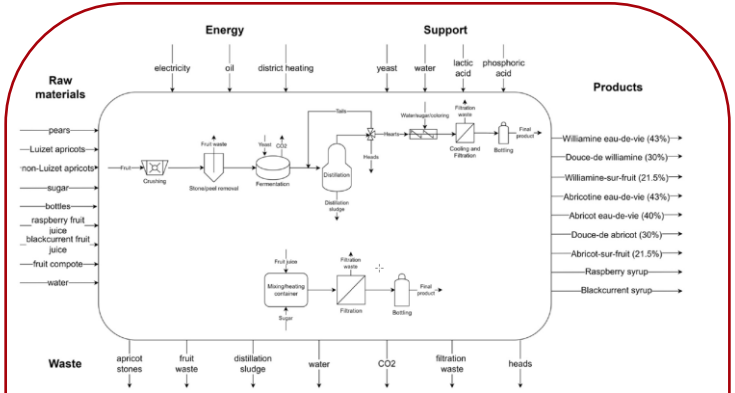
MAS audits database

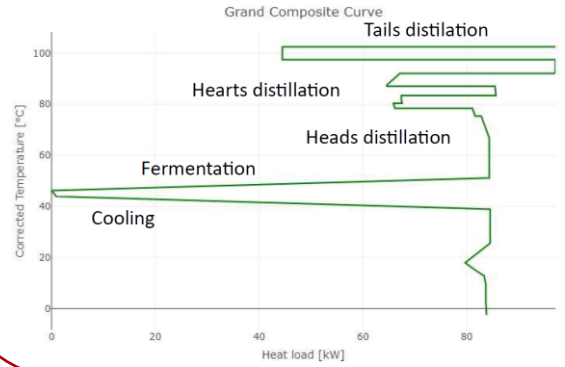
Explosives plant



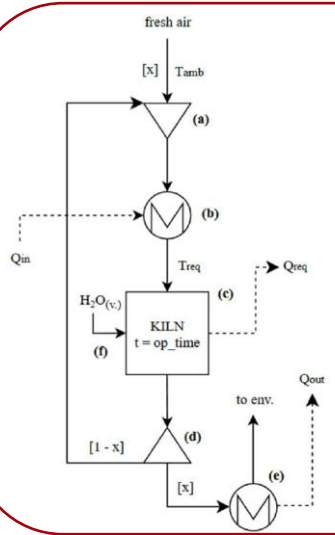


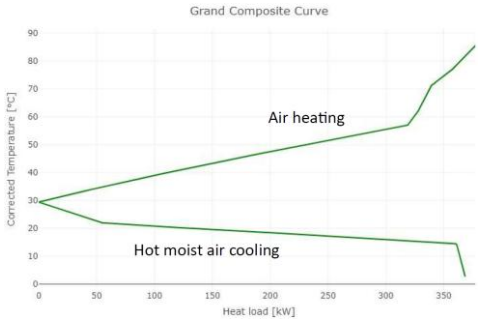
Distillery plant





Malting plant



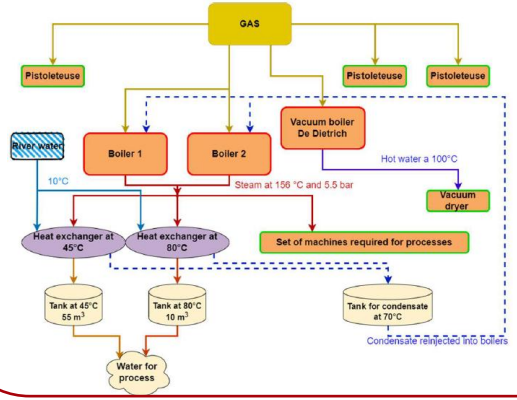


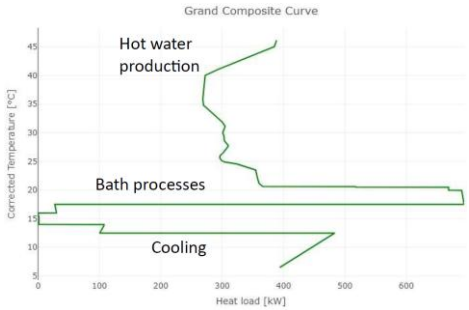
Florez-Orrego et al (2024) Swiss OFEN. https://pubdb.bfe.admin.ch/en/publication/download/11946

https://gitlab.epfl.ch/ipese/mas/masbook

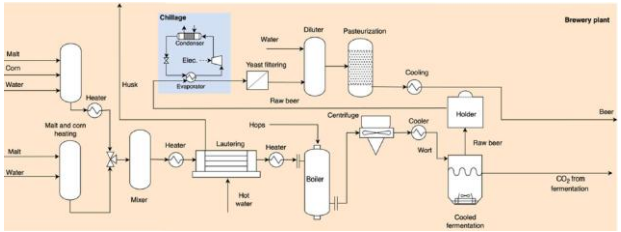
MAS audits database

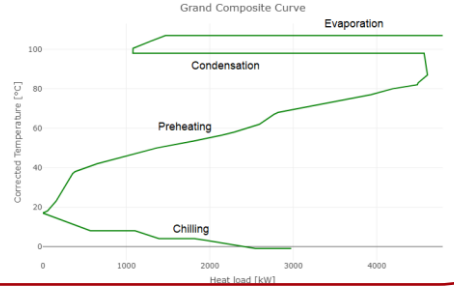
Tannery model



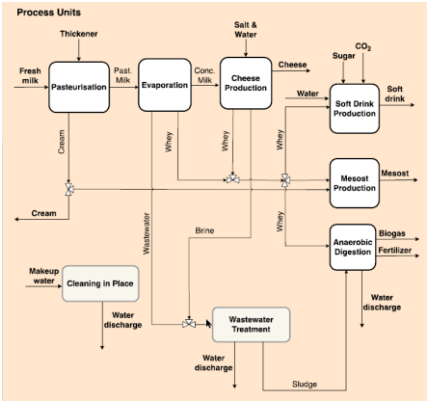


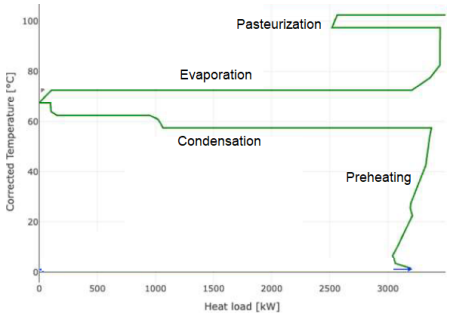
Brewery model





Dairy model





Florez-Orrego et al (2024) Swiss OFEN.
https://pubdb.bfe.admin.ch/en/publication/download/11946
https://gitlab.epfl.ch/ipese/mas/masbook